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Recent developments in the Last-Mile Location-Routing problem

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ABSTRACT

Recent technological advancements have facilitated various initiatives in the last-mile location routing problem (LM-LRP), yet existing LM-LRP literature largely evaluates these initiatives in isolation and lacks a unified framework. This study aims to address this limitation by conducting a systematic literature review of emerging initiatives in LM-LRP to synthesize their technological and operational implications. First, transportation nodes and modes are categorized to show how emerging initiatives change the spatial configuration of logistics facilities and affect sustainability dimensions, while increasing coordination requirements across LM-LRP networks. The literature is then assessed based on key LM-LRP features, including distribution network structure, objective-function design, and last-mile applications. A digital twin perspective is further proposed as a unified decision-support system to connect LM-LRP components with data-driven modeling and simulation approaches. Sustainability trade-offs are subsequently examined across emerging LM-LRP initiatives, showing that their economic, environmental, and social benefits remain context-dependent. Key research gaps are identified, including limited operational realism, fragmented technology integration, and insufficient governance and policy considerations. Based on these findings, a forward-looking framework is proposed to summarize initiative adoption and future development pathways. Finally, this review provides actionable research insights that can facilitate the implementation of emerging initiatives for more efficient last-mile distribution networks.

1. Introduction

The last-mile location-routing problem (LM-LRP) has been rapidly reshaped by technological advancements and digitalization. The LM-LRP business environment has become increasingly competitive, mainly due to customers' intense pressure for faster and cheaper delivery and the surge in the number of last-mile logistics firms (Amini & Haughton, 2023; Rautela et al., 2022). In such a market, one way to stay competitive is to redesign logistics networks to increase efficiency and profitability (Ozyavas et al., 2025). This process is increasingly enabled by recent technological advances, such as the use of new transportation nodes like parcel lockers (PLs) for demand fulfillment (Ozyavas et al., 2025) or advanced transportation modes, e.g., autonomous delivery robots (ADRs) for last-mile delivery (Ghiani et al., 2025). Additionally, integrative concepts, e.g., the Physical Internet (PI), promote open and shared networks (Liu et al., 2022), thereby catalyzing the implementation of emerging initiatives in LM-LRP.

Despite the studies conducted to examine the impact of node and mode initiatives, the majority of them focus solely on nodes or modes. This fragmentation causes the designed network to be suboptimal, thereby increasing inefficiency in the last mile and leading to the underutilization of the introduced initiatives (Zhao et al., 2025). This limitation is critical as it increases total cost and reduces operational feasibility. For example, incorporating ADR or drone delivery modes into the network while assuming that human operators handle loading and unloading, even though some nodes, such as PLs, enable fully autonomous loading and unloading, reflects a lack of integration. On the other hand, understanding when and where to employ advanced nodes and modes is closely tied to the structure of the LM-LRP. This structure is fundamentally shaped by the network's echelon configuration, a discrete hierarchical level defined by a specific class of nodes where inventory is stored, processed, or aggregated before being transferred to the next stage (Snyder & Shen, 2007). However, the integration of emerging node and mode innovations and their positioning within the distribution network are lacking in the literature.

The adoption of emerging technologies in the LM-LRP faces *three interrelated problems*.

First, the existing literature lacks a unified framework that integrates

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both established and emerging transportation nodes and modes within the broader LM-LRP network structure, leading to fragmented and often suboptimal designs. Second, insufficient governance mechanisms limit collaboration and information sharing among stakeholders, reducing the practical adoption of shared nodes and collaborative delivery concepts. Third, incorporating new initiatives increases system complexity, a challenge that is particularly pronounced for transportation mode-related technologies, where operational feasibility depends on tight coordination, synchronization, and infrastructure compatibility.

Governance barriers to the adoption of emerging initiatives in the LM-LRP hinder, or at least delay, their practical implementation (Rautela et al., 2022). For example, open and shared-access transportation nodes, such as PLs and PI-hubs, have been introduced to distribute opening costs across multiple users and increase utilization rates by serving aggregated demand (Kahalimoghadam et al., 2025a). Although these facilities enhance efficiency, logistics companies often limit their participation in collaborative networks due to competitive concerns and a reluctance to disclose sensitive operational information.

Moreover, incorporating new initiatives into the LM-LRP increases system complexity, a challenge that is particularly pronounced for technologies related to *transportation modes* (Bruni et al., 2023). ADRs, drones, and self-driving vehicles are automated modes that can support continuous operation and facilitate electrification goals. However, limited battery capacity restricts their operating range, increases charging frequency relative to conventional vehicles, and reduces payload capacity (Kitjacharoenchai et al., 2020; Li et al., 2021a). Multimodal designs, such as truck–drone systems (Kitjacharoenchai et al., 2020; Ouhader & El Kyal, 2023), van-bike and van-walk letter delivery (Kahalimoghadam et al., 2026), and van–robot systems (Yu et al., 2022), as well as integrations with public transport, have been proposed to address the limitations of electric modes; however, they further increase system complexity, as their effectiveness relies on precise intermodal coordination and synchronization. Accordingly, prior work has expanded node and mode functionalities but has not yet consolidated how they should be jointly positioned, governed, and operationalized within technology-driven LM-LRP networks.

To address these gaps, this study conducts a comprehensive *systematic literature review* to synthesize technological advances across transportation nodes, modes, and network structures in the LM-LRP. The systematic literature review method is selected because it provides an evidence-based synthesis of high-quality studies in the field and minimizes bias by ensuring transparency and rigor in the selection of the reviewed articles (Lame, 2019). This systematic literature review structures fragmented knowledge in the LM-LRP by revealing adoption patterns and mapping relationships among emerging initiatives. In doing so, the review proposes a digital twin perspective that not only connects physical LM-LRP components with data-driven modeling but also integrates various modeling and simulation approaches to create a unified decision-support system. Additionally, it provides a structured synthesis of technology-specific gaps, conflicting findings, and sustainability trade-offs across economic, environmental, and social dimensions. This synthesis supports the development of a practical roadmap that illustrates the current state of technology adoption and identifies the actions required in the near future.

Several well-recognized literature reviews on the LRP have been published, including the studies by Prodhon and Prins (2014) and Drexl and Schneider (2015). However, their coverage ended nearly a decade ago and was limited to the classical LRP, which constrains their relevance for technology-driven LM-LRP developments. Moreover, although recent review studies, such as Winkelhaus and Grosse (2019), Lagorio et al. (2020), and Chung (2021), addressed technological innovations at the logistics system level, they did not analyze how such innovations affect LM-LRP modeling decisions or network structure evolution. Thus, the implications for last-mile design remain insufficiently explored. Therefore, the contributions of this study to the LM-LRP literature are highlighted below.

First, we synthesize LM-LRP research by systematically classifying transportation nodes, modes, and integrative paradigms, linking them to echelon structures and modeling decisions, and clarifying their modeling requirements and practical adoption challenges. Each study is systematically evaluated against the main LM-LRP applications and characteristics, including single- or multi-objective formulation, representation of uncertainty, stochastic programming, sustainability assessment, and synchronization. Second, we structure how the emergence of new initiatives has shifted LM-LRP structures toward more complex systems and map these initiatives and their interactions in both the real and digital worlds. The interdependencies between the primary objectives of the LM-LRP and the enabling roles of node and mode initiatives are uncovered. Finally, we translate the identified gaps into a forward-looking framework that defines concrete research directions for advancing LM-LRP under emerging initiatives. Solution methodologies used in the LM-LRP are classified in Appendix A, and their applicability across different nodes, modes, and network structures is discussed.

The remainder of the paper proceeds as follows. Section 2 describes the systematic literature review protocol. Section 3 examines node- and mode-related trends in the LM-LRP and classifies the integrative concepts that have emerged in this domain. Section 4 categorizes the LM-LRP studies by distribution network structure. Focusing on the current operational landscape of the LM-LRP, Section 5 synthesizes the operational trade-offs arising from different node and mode initiatives. Section 6 identifies key gaps in the LM-LRP and introduces an integrated framework with five development areas and five generations. This framework acts as a roadmap for future empirical and methodological research. Section 7 concludes the paper. Finally, Table 1 presents the list of abbreviations defined in this study.

2. Literature review method and bibliometric overview

2.1. Systematic literature review methodology

Fig. 1 presents the methodology of the conducted systematic literature review and summarizes the paper selection process. This literature review methodology is designed in accordance with the preferred reporting items for systematic reviews and *meta*-analyses (PRISMA) 2020 guidelines (Page et al., 2021) to ensure a fair and transparent search strategy, explicit inclusion and exclusion criteria, and clear reporting of the study selection and screening process. After scoping the research domain, Scopus and Web of Science were used to retrieve peer-

Table 1
List of abbreviations used in this study.

Abbreviation	Definition
1E-LRP	A single echelon LRP
2E-LRP	Two-echelon LRP
2E-VRP	Two-echelon vehicle routing problem
ADMM	Alternating direction method of multipliers
ADRs	Autonomous delivery robots
ALICE	Alliance for logistics innovation through collaboration in Europe
ALNS	Adaptive large neighborhood search
CO_2	Carbon dioxide
EVs	Electric vehicles
GPS	Global positioning system
IoT	Internet of Things
LNS	Large neighborhood search
LM-LRP	Last-mile location routing problem
MILP	Mixed-integer linear programming
MIP	Mixed-integer programming
NSGA-II	Non-dominated sorting genetic algorithm II
PI	Physical Internet
PLs	Parcel lockers
PRISMA	Preferred reporting items for systematic reviews and *meta*-analyses
UCCs	Urban consolidation centers
VNS	Variable neighborhood search
WMA	Weighted moving average

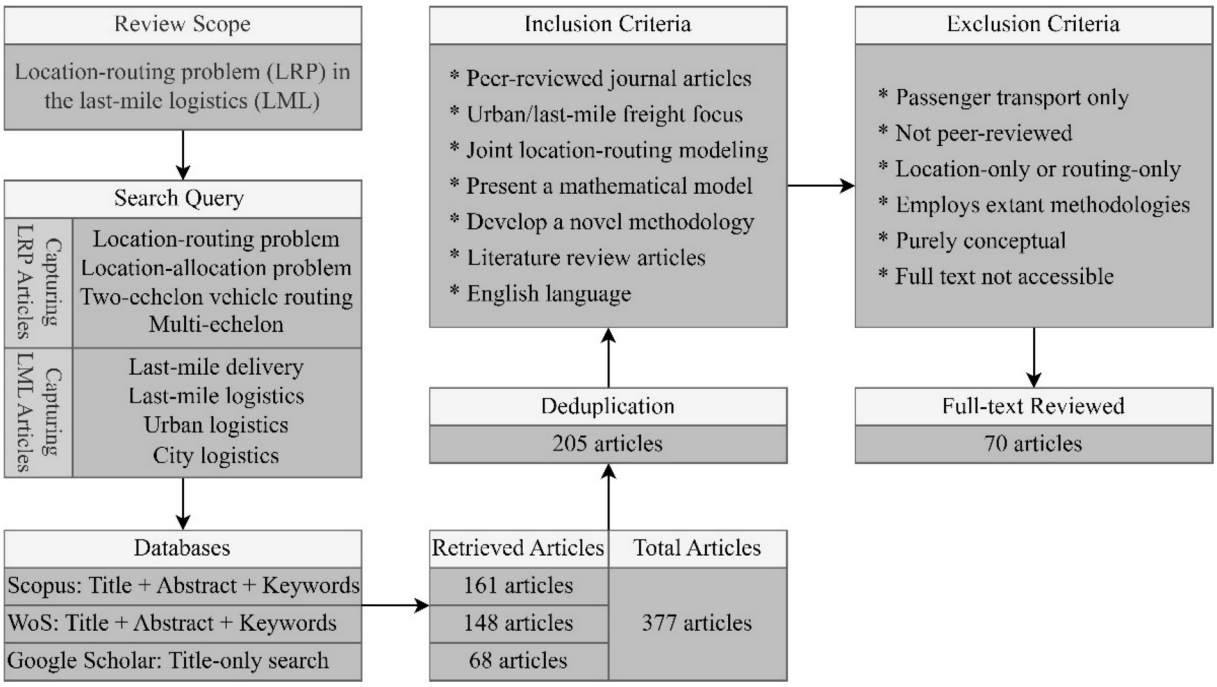


Fig. 1. Methodology for the systematic literature review on LM-LRP.

reviewed articles, while Google Scholar was used to broaden coverage. This approach aligns with the methodological guidance highlighted by Niforatos et al. (2019) who emphasize that effective search strategies should combine controlled vocabulary with free-text terms to minimize the risk of missing relevant studies. After that, a Boolean structure was designed to reflect the topic through conceptual blocks and maximize search transparency (Caulley et al., 2020).

In Scopus and Web of Science, structured Boolean queries across titles, abstracts, and keywords were applied to capture English-language, peer-reviewed journal articles relevant to LM-LRP. This query was: TITLE-ABS-KEY ((“Location-routing problem” OR “location routing” OR “hub location and routing” OR “location-allocation problem” OR “two-echelon vehicle routing” OR “multi-echelon system”) AND (“last-kilometer delivery” OR “last-kilometer logistics” OR “last-mile” OR “last-mile delivery” OR “last-mile logistics” OR “last-kilometer” OR “urban logistics” OR “city logistics”)).

In Google Scholar, due to its search limitations, which do not allow filtering within abstracts or keywords, search queries were applied across the full text, including article references, which can reduce the relevance of retrieved studies to the research topic. Therefore, we restricted the search to title-only queries to improve result precision, although this resulted in a smaller pool of studies. The revised search query in Google Scholar was intitle:(“location-routing” OR “location routing” OR “location-allocation” OR “hub location” OR “two-echelon” OR “multi-echelon”) AND intitle:(“last-mile” OR “last-kilometer” OR “urban logistics” OR “city logistics”). Scopus returned 161 records, Web of Science 148, and Google Scholar 68. In total, 377 articles were captured, which were imported into EndNote to remove duplicates across databases. Additionally, the search was conducted in July 2025, with no restriction on publication year. After deduplication, 205 unique articles remained, covering the period 2003–2025.

The final article refinement step was to apply inclusion and exclusion criteria to ensure that relevant and rigorous studies were included in the review. Eligible studies were English-language, peer-reviewed journal articles that examined urban freight, last-mile, or city-logistics applications and offered clear contributions to the LM-LRP, including models that addressed both location and routing decisions together. Another inclusion criterion was that all studies, except literature reviews, had to include a mathematical model. This helped ensure that the selected papers provided more than general discussion and included a clear method for the LM-LRP. Exclusions applied to studies with topical mismatches, such as those addressing location without routing, routing without location, or passenger-only contexts with no clear freight relevance. Purely conceptual work without an applied methodology or a coherent motivating case was also excluded, unless it offered substantial theoretical advances with direct implications for urban freight decision making.

After applying the inclusion and exclusion criteria, 70 papers were retained for full review. Among these, 65 articles were published in Q1 journals, with the remainder appearing in Q2 journals. This strong presence of high-tier publications adds confidence to the overall quality of the review and to the insights drawn from it. It also allows gaps in the literature to be identified more clearly, helping to steer future research toward practical and implementable solutions.

2.2. Publication distribution

In this section, the overall statistics and trends regarding publication distributions are presented. Fig. 2 shows the number of published papers per year alongside the 3-year weighted moving average (WMA), indicating sustained growth in LM-LRP publications. Publications were

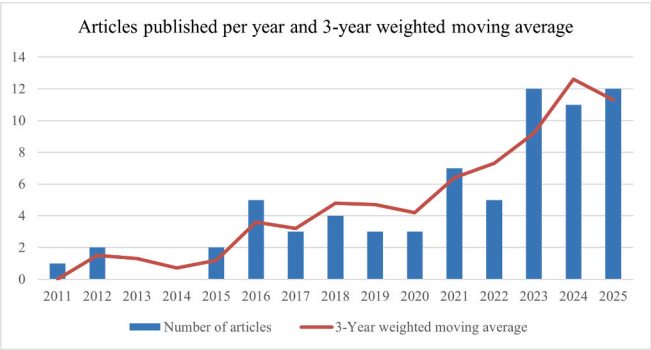


Fig. 2. Annual LM-LRP publications and three-year WMA, 2011–2025.

sparse in the early 2010 s, increased from 2016, and accelerated after 2021, with peaks in 2023 and 2024. The mild easing in 2025 reflects year-to-date counts, as the search ended in July 2025. The 3-year WMA increases from about one to more than eleven, indicating sustained growth. As shown in Fig. 3, publication outlets are concentrated in core operations research and transportation journals. The top five journals together represent about 54 percent of the 70 papers, while the top ten account for roughly 74 percent.

3. Trends in the LM-LRP

This section synthesizes the LM-LRP into its core components, nodes and modes, which correspond to location and routing, and then surveys the integrative concepts identified in the field. It evaluates how recent node archetypes and mode innovations reshape urban distribution, the modeling opportunities they create, and the practical challenges associated with adoption.

3.1. Transportation node trends in the LM-LRP

Historically, LM-LRPs determined the number, capacity, and location of facilities, which suited stable demand and long lead times. In contrast, last-mile systems face e-commerce-driven, high-frequency, low-volume demand, while urban constraints, such as congestion and low-emission zones, further reduce operational efficiency. These pressures are pushing transportation nodes closer to customers to shorten final-mile segments and expand coverage, while preserving flexibility for spatial and temporal variability. This approach leads to the emergence of new node initiatives.

The most adopted nodes in the LM-LRP are urban consolidation centers (UCCs), which function as intermediary hubs strategically situated close to or within the periphery of urban areas. They accept deliveries from numerous warehouses via heavy-duty transport, consolidate shipments, and subsequently distribute them to final customers using smaller, typically low-emission vehicles (Juvvala & Sarmah, 2025; Simoni et al., 2018). However, the economic sustainability of UCCs has been criticized by many studies. A viable solution is the shared use of these hubs, not only by spreading fixed costs but also by increasing vehicle load factors as more parcels become available (Ouhader & El Kyal, 2023). Another challenge is synchronization between inbound and outbound vehicles, especially when delivery time windows require more comprehensive modeling procedures (Simoni et al., 2018).

Other intermediate and customer-inaccessible nodes are satellites, where consolidation and goods transfer occur. Goods from warehouses in the outer echelon are transferred to these nodes for final delivery. Unlike UCCs, collaboration usually does not happen at satellites, as they are owned by a single company, either a shipper or a carrier. Satellite size is usually small, enabling multiple sites within the distribution network, which raises drop density and shortens last-mile routes (Anderluh et al., 2021; Breunig et al., 2016; Zhou et al., 2024). Accordingly, they are located between the intermediate and inner echelons, which causes higher opening costs. This issue can be handled by balancing opening and operational costs through reducing total vehicle-kilometers traveled (Du et al., 2023). Lower opening costs can be achieved by using mobile satellites, as they are opened when and where they are needed, although re-optimization is required (Liu et al., 2021; Pourmohammad-Zia & van Koningsveld, 2024; Sutrisno & Yang, 2023; Tian & Hu, 2023; Yu et al., 2020). Finally, synchronization is a challenge when implementing satellites, as the problem usually involves time windows, goods handover between modes, and load-dependent travel times.

Dark stores are customer-inaccessible micro-fulfillment facilities, typically repurposed from retail or small warehouse spaces, and designed for rapid order picking and click-and-collect services. Within the LM-LRP formulations, they act as proximate fulfillment nodes that complement larger depots by reducing last-mile distances, increasing drop density, and improving responsiveness for time-sensitive e-commerce deliveries (Tahirov et al., 2025). Their operations are characterized by constrained storage, short picking cycles, and frequent replenishment. From a modeling perspective, LM-LRPs capture the trade-off between the fixed and operating costs of opening dark stores and the potential reductions in routing time, vehicle use, and service delays. Economic feasibility is further shaped by factors such as secure site access, efficient picking labor, and reliable replenishment scheduling.

Micro-hubs, or micro-consolidation centers, are compact versions of UCCs designed for inner urban areas. Their functions include consolidation, sorting, temporary storage, and online demand fulfillment (Kahalimoghadam et al., 2024). Parcels or fresh food are transferred to micro-hubs by medium trucks or vans and then delivered to customers by cargo bikes or ADRs, with customer pickup also possible (Liu et al., 2022; Rave & Fontaine, 2025). Micro-hubs, positioned between intermediate and inner echelons, enhance e-commerce agility. However, being closer to inner urban areas entails higher opening costs and greater exposure to lease price uncertainty. These challenges are addressed through pooled replenishment strategies that share costs among stakeholders (Guo et al., 2024) and robust optimization approaches that account for demand and operational uncertainty in the LM-LRP models (Liu et al., 2022).

PI-hubs are standardized, open interchange nodes that decouple freight flows from last-mile service by introducing a hub-and-spoke layer. They can be viewed as shared versions of UCCs or micro-hubs and accommodate multiple stakeholders, typically shippers and carriers. By pooling demand, PI-hubs absorb volume fluctuations more effectively than traditional nodes and support both vertical and horizontal collaboration. Inbound goods are handled by trucks or vans, while final delivery is undertaken by smaller, low-emission modes such as electric vans, cargo bikes, and ADRs. In the LM-LRP formulations, PI-hubs introduce additional elements including synchronization requirements, heterogeneous fleets, and PI-usage fees (Liu et al., 2021). Stakeholder interactions can also be represented through a multi-agent system, where behavior is modeled in a learning environment using reinforcement learning supported by fair reward structures (Kahalimoghadam et al., 2025b).

PLs are unattended pickup stations where multiple carriers deposit parcels for later customer collection, serving as consolidation and transfer nodes. They may be fixed or mobile and are typically placed at high-access sites, e.g., shopping centers and stations. Mobile variants can be repositioned to follow demand and time-window patterns under uncertainty (Enthoven et al., 2020; Ozyavas et al., 2025). Other advantages of PLs include reduced failed deliveries and shorter last-mile legs. They can also be paired with cargo bikes or electric vehicles (EVs) for replenishment and collection, which enhance sustainability performance (Mirzavand Boroujeni et al., 2025; Zhao et al., 2025). PLs can be shared by multiple carriers, which increases utilization rates and reduces total costs, but also increases operational complexity as it requires explicit capacity representation (Korkmaz et al., 2025; Ozyavas et al., 2025; Zhao et al., 2025). Finally, PLs are also well suited for integration with automated delivery modes, such as ADRs and drones, enabling automated parcel transfer without human intervention.

Table 2 provides clear definitions of all nodes identified in the LM-LRP, enabling easy comparison for future modeling purposes. In addition, a conceptual diagram is developed for each node to visualize where it can be implemented within the distribution network. Although the diagrams illustrate three echelons, outer, intermediate, and inner echelons, their primary purpose is to indicate the most suitable location for each node in the urban landscape and its proximity to city centers. A distribution network may therefore operate with only one echelon and still incorporate all nodes. Modeling implementation criteria are also summarized in Table 2, specifying the type of network on which each node can be used and the key decisions that need to be made in the

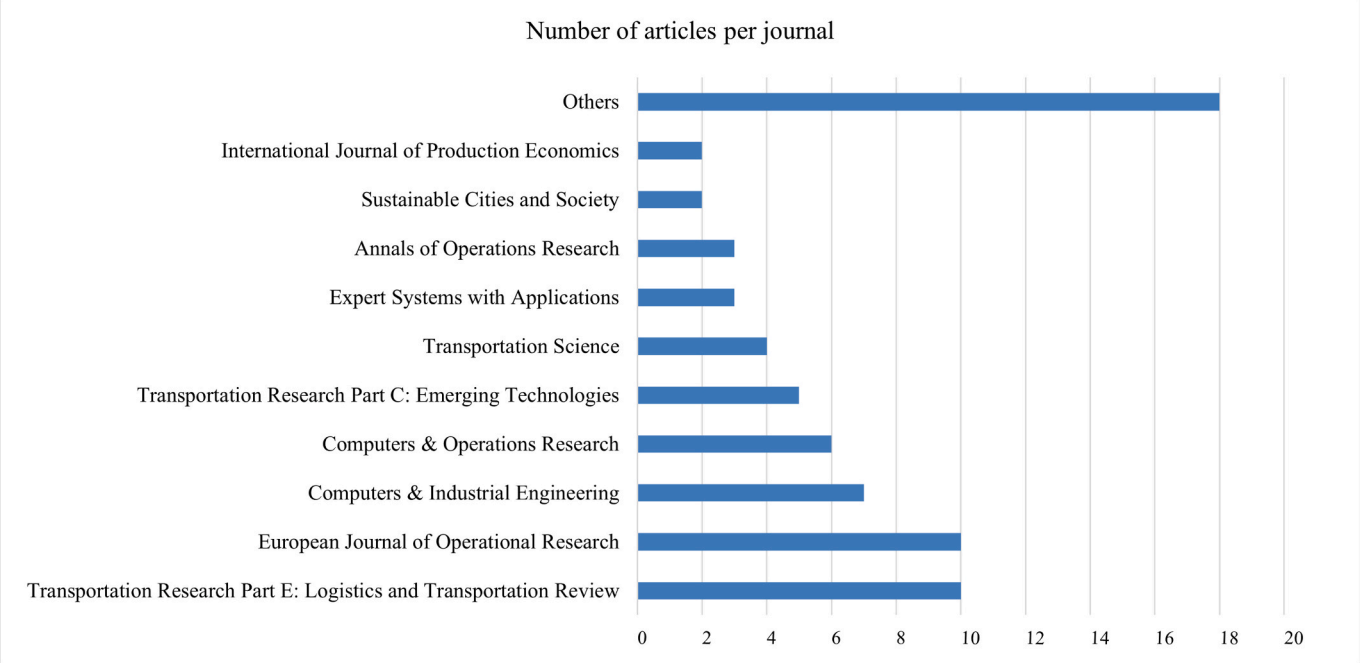


Fig. 3. Number of articles published per journal. Single-paper journals are grouped as "Others", with the full list provided in Appendix B.

modeling process. Finally, the advantages and disadvantages are listed to provide a realistic view of the benefits that can be achieved and the challenges that may arise when integrating these nodes into an LM-LRP.

Fig. 4 shows the chronological evolution of node initiatives in the LM-LRP literature. Satellites appear consistently across the review period and show a marked increase after 2020, reaching a peak in 2023, before declining in 2024–2025. PLs have gained increasing attention since 2019, while UCCs show intermittent but recurring adoption. Micro-hubs are more recent and remain less frequent, with studies appearing mainly after 2022.

3.2. *Transportation modes trends in LM-LRP*

Transportation modes are one of the main sources of emissions and air pollution in urban areas. They contribute to 23% of emissions, of which 41% arises from freight transport, corresponding to approximately 2.8 gigatons of carbon dioxide (CO_2) (Fry et al., 2024; Zhang et al., 2024). However, environmental awareness and the rise of actionable climate policies, such as the Sustainable Development Goals and the Paris Agreement, have led to several climate initiatives, including zero- or low-emission zones (Kahalimoghadam et al., 2022). On the other hand, current technological advancements, especially improvements in batteries, have increased the popularity of electric modes, from small modes, e.g., ADRs and drones, to medium and large modes, e.g., electric vans and electric trucks. This section examines the range of emerging and established transportation modes currently used in the LM-LRP and compares them to provide a comprehensive overview of their benefits and drawbacks.

EVs represent one of the most widely adopted initiatives with the potential to substantially electrify the LM-LRP. To achieve a green and modern network, many studies have evaluated the combination of EVs with PLs and home deliveries in inner echelons (Korkmaz et al., 2025; Mirzavand Boroujeni et al., 2025; Schneider & Löffler, 2017) or replaced conventional vans and waterborne transport modes serving UCCs and micro-hubs with EVs (Juvvala & Sarmah, 2025; Pourmohammad-Zia & van Koningsveld, 2024). Incorporating EVs into the LM-LRP increases modeling complexity, as the model must account for state-of-energy constraints, temperature-dependent energy consumption, charging or swapping options with partial charges, nonlinear charging curves, and station-capacity limits (Chen et al., 2024b; Janinhoff et al., 2024; Leng et al., 2023; Yang et al., 2022). Charging station locations, battery aging, and the total cost of ownership are additional features that must be examined to encourage stakeholders to adopt electrification (Du et al., 2023; Leng et al., 2023; Wang et al., 2024; Wang et al., 2023; Yang et al., 2022).

Cargo bikes are one of the zero-emission modes that can be used in the inner-echelons areas with high customer density. Multiple studies have evaluated their viability, especially in collecting goods from PLs, satellites, or mobile hubs and delivering them to customers (Enthoven et al., 2020; Rave & Fontaine, 2025). Coverage radius, maximum payload, cycling infrastructure, and cycling speed are among the cargo bike constraints that must be included in the LM-LRP. Cargo bikes are highly compatible with densely populated inner echelons and areas subject to low- or zero-emission zoning restrictions. Moreover, they can significantly improve social sustainability, as the fatal accident rate caused by cargo bikes is very low (Anderluh et al., 2021; Ozyavas et al., 2025; Simoni et al., 2018).

ADRs are another mode initiative,[1] which are electrically powered and ground-based modes that operate on pedestrian and curbside paths to perform short-range parcel and grocery deliveries. They are most suitable for dense inner-echelon environments characterized by short service radii and small payload requirements. Key characteristics that must be represented in LM-LRP models include their low speeds, limited battery range, and restricted payload capacity. These constraints mean ADRs are often used in allocation-type problems, typically collecting single-customer parcels from public transport nodes, micro-hubs, or PLs and returning for reloading or charging (Liu et al., 2022; Yu et al., 2022). They can also be deployed through a van–mothership strategy, where vans carry ADRs, release them at designated stop points, and later collect them at the same or another location. Such multimodal configurations require tight synchronization, especially in LM-LRPs with time windows. Despite limitations related to weather operability and regulatory conditions, ADRs are highly effective in congested urban areas

[1] Terminology varies across studies and regions, with labels such as driverless delivery robots, ground drones, or van-launched micro-hub robots (Ghiani et al., 2025; Li et al., 2021a).

Table 2
Key nodes in the LM-LRP and their roles in the network. The color gradient indicates each node's typical location in the distribution network. (Dark blue: higher feasibility; lighter shades: lower feasibility).

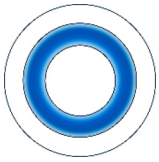
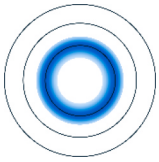
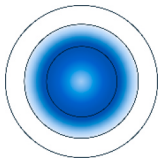
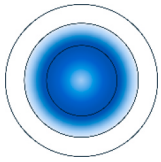
Node	Definition	Modeling Implementation	Advantages / Disadvantages
UCCs 	UCCs serve as intermediate consolidation points at or near the urban boundary. They take freight from multiple suppliers by heavy-duty vehicles, consolidate it, and forward it to customers with smaller, generally low-emission vehicles.	– Two-echelon and multi-echelon LM-LRP – Decision: opening and operational costs, capacity, customer allocation, fleet composition, long time windows and synchronization	✓High drop density ✓Short last-mile legs ✓Promotes green logistics ×High facility costs ×Underutilization risk ×Governance and coordination complexity ×Need public support to be economically sustainable ×Not accessible for pickup
Satellites 	Located on the periphery or within urban areas, a satellite is an intermediate distribution point that receives goods from a central depot. It supports order consolidation, temporary storage, and dispatches goods to final destinations.	– Two-echelon and multi-echelon LM-LRP – Decisions: opening and operational costs, capacity, customer allocation, fleet composition, long time windows, and synchronization	✓Short last-mile legs ✓Enhanced delivery speed ✓Scalability and support decentralized logistics ×Not accessible for pickup ×Site selection constrained by land-use rules
Dark stores 	Customer-inaccessible micro-fulfillment nodes in urban or *peri*-urban areas, often converted from retail or small warehouse spaces, designed to support rapid order delivery and click-and-collect operations.	– Two-echelon and multi-echelon LM-LRP – Decisions: opening and operational costs, storage and capacity, inventory level, customer allocation, service radius, and short time windows	✓High drop density ✓Shorter last-mile legs ✓Agile delivery and click-and-collect ×Higher facility costs ×Site selection constrained by land-use rules
Micro-hubs 	Compact transshipment facilities located within urban areas, where parcels arrive via medium-small modes and are transferred to lighter modes for final delivery. They decentralize service and shorten last-mile segments.	– Two-echelon and multi-echelon LM-LRP – Decisions: opening and operational costs, capacity, customer allocation, service radius, fleet composition, and short time windows	✓High drop density ✓Shorter last-mile legs ✓Agile delivery and click-and-collect ✓Promotes green logistics ×Higher facility costs ×Risk of underutilization ×Site selection constrained by land-use rules

Table 2 (*continued*)

Node	Definition	Modeling Implementation	Advantages / Disadvantages
PI-hubs 	PI-hubs are standardized, modular, shared nodes that consolidate, sort, and transship parcels. They facilitate vertical and horizontal collaboration by enabling the dynamic exchange of goods across connected logistics networks with interoperable, standardized units.	– Two-echelon and multi-echelon LM-LRP – Booking system and auction for services – Decisions: opening and operational costs, dynamic and real-time storage and capacity, customer allocation, service radius, and short time windows	✓Very high drop density ✓Facility cost reduction enabled by sharing ✓Agile delivery and click-and-collect ✓Promotes green logistics ✓Lower underutilization risk ×Site selection constrained by land-use rules ×Higher complexity due to shared network
PLs 	PLs are automated and unattended pickup stations where multiple carriers deliver parcels for customer collection. They can be fixed or mobile and act as transfer and consolidation points and are typically placed at accessible locations, e.g., shopping centers and stations.	– Two-echelon and multi-echelon LM-LRP – Booking system – Decisions: opening and operational costs, dynamic and real-time capacity, customer allocation, service radius, and short time windows	✓High drop density ✓Low facility cost due to small size ✓Agile delivery and click-and-collect ✓Promotes green and active mode logistics ×Limited capacity ×Less attractive to customers if only pickup is available

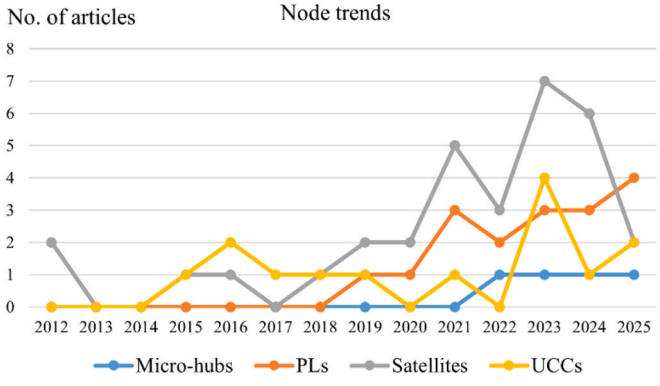


Fig. 4. Chronological trend of node initiatives in LM-LRP studies.

and zero-emission zones (Ghiani et al., 2025; Liu et al., 2022).

Drones are fully electric, autonomous aerial delivery modes incorporated into the LM-LRP applications. Compared with ADRs, drones offer greater range, higher speeds, and larger effective payloads, which translate into broader operational applicability in the LM-LRP. They are used for allocation deliveries, mothership deliveries (usually by trucks), and deliveries to remote areas (Bruni et al., 2023; Kitjacharoenchai

et al., 2020; Tian & Hu, 2023). Their advantages include bypassing surface congestion, achieving faster travel times, and enabling fully automated pickup and delivery, particularly in networks integrated with PLs. However, compared to conventional modes, drones pose notable modeling challenges due to limited payload and battery capacity (Chen et al., 2024a; Ghiani et al., 2025). As the state of the battery is highly dependent on payload and flying conditions, battery usage, and therefore travel range, should be considered uncertain parameters, necessitating robust and stochastic programming (Korkmaz et al., 2025). Public acceptability and regulatory approval for drones are other challenges that must be addressed.

The discussed mode initiatives can be integrated with conventional delivery modes within a multimodal LM-LRP. An example of this multimodal adaptation is the use of conventional vehicles, e.g., vans and trucks, as motherships for drones and ADRs (Kitjacharoenchai et al., 2020). Synchronization between modes and finding optimal points for deploying and picking up autonomous modes are additional constraints in multimodal LM-LRPs (Bruni et al., 2023; Korkmaz et al., 2025; Liu et al., 2022). In return, multimodal strategies enable congestion avoidance, reduced service times, and the potential for significant cost savings when employing ADRs and drones (Chen et al., 2024a; Mirzavand Boroujeni et al., 2025). Table 3 summarizes key performance indicators for the identified mode initiatives in the LM-LRP and highlights their deployment conditions.

The mode-related trend shown in Fig. 5 illustrates a clear increase in technology adoption after 2020. EVs are the most frequently represented mode initiative, with a steady upward pattern and the highest value in 2025. Drones show limited presence before 2021 but become more visible from 2023 onward. Cargo bikes appear intermittently, with a noticeable peak in 2023 and renewed attention in 2025. Finally, ADRs remain less frequent overall, with limited but recurring representation after 2021.

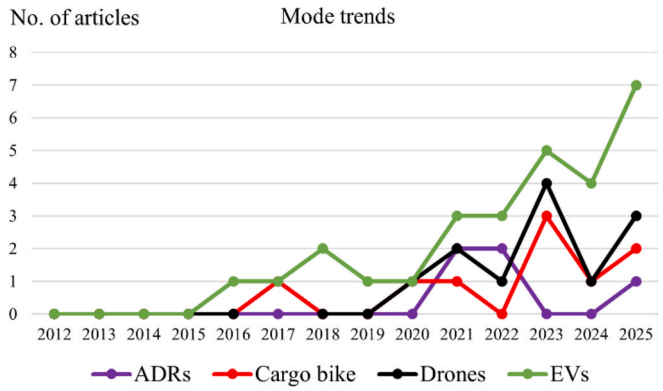


Fig. 5. Chronological trend of mode initiatives in LM-LRP studies.

3.3. Integrative concepts in the LM-LRP

This section outlines key concepts shaping the LM-LRP. It first examines the roles of City Logistics and the PI in advancing sustainability through coordinated urban planning and resource sharing and then introduces digital twin technologies as data-driven enablers that unify these paradigms.

3.3.1. City logistics

City Logistics coordinates urban freight across infrastructure, land use, regulation, and operations to improve reliability while reducing congestion, emissions, and risk. In the LM-LRP, it promotes consolidation nodes, such as UCCs, satellites, and micro-hubs, to pool upstream goods, then transfer them to customers. This strategy reduces costs and emissions by balancing opening and operational costs (Hemmelmayr et al., 2012; Juvvala & Sarmah, 2025). City Logistics also enhances environmental sustainability by promoting emission reduction policies, such as low-emission zones, and supporting heterogeneous fleet composition that enables the deployment of EVs (Koç et al., 2016). Finally, emphasizing collaborative LM-LRP models to aggregate demand is a complementary concept to node initiatives, such as UCCs, which are financially viable when demand volume is high (Zhao et al., 2018).

Table 3
Advantages and disadvantages of key modes in the LM-LRP. The color gradient indicates each mode's typical location in the distribution network. (Dark blue = higher feasibility; lighter blue = lower feasibility).

	Emissions	Cost	Load	Speed	Echelon
EVs	✓ Zero emissions – fuel efficient	✓ Low cost, particularly in low emissions zones	✓ High – suitable for routing	✓ High but not suitable for congested area	
Cargo bikes	✓ Zero emissions	✓ Very low cost	= Medium – suitable for routing	= Medium - suitable for congested area	
ADRs	✓ Zero emissions - need frequent charging	✗ Medium–high cost, needs optimisation	✗ Low – suitable for allocation	✗ Low but suitable for congested area	
Drones	✓ Zero emissions - need frequent charging	✗ Medium–high cost, needs optimisation	✗ Usually low – suitable for allocation	✓ High - suitable for congested area	
Conventional vans	✗ Produce emissions	= Low cost but excluded from low-emission zones.	✓ High – suitable for routing	✓ High but not suitable for congested area	

3.3.2. Physical Internet

The PI is an open, standardized, and interoperable logistics paradigm that enables modular containers to be dynamically consolidated and routed across shared networks using standardized interfaces and protocols. In the LM-LRP context, the PI induces a shared, multi-echelon architecture that consolidates flows at micro-hubs and coordinates mixed fleets, with heavier modes line-hauling from depots to hubs and lighter, and more environmentally friendly modes completing last-mile service. In a PI-enabled last-mile network the efficiency of modes and nodes is enhanced by consolidating flows at shared hubs, coordinating capacity across users, and enabling multimodal transfers (Liu et al., 2022). The higher consolidation level and load factor directly reduce the total number of modes and, consequently, vehicle kilometers traveled. Hence, PI-enabled networks reduce total costs and emissions in LM-LRP.

3.3.3. Digital twins

A digital twin is a high-fidelity, lifecycle-spanning, continuously synchronized virtual replica that fuses data, models, and operational constraints to monitor, predict, and actuate logistics in real time. In the LM-LRP, a digital twin represents nodes, modes, routes, schedules, and external constraints, such as construction, incidents, traffic conditions, and access rules, as observable system states updated through telemetry and infrastructure feeds. This representation enables rapid what-if analysis, anticipatory re-optimization, and closed-loop interventions as demand, capacity, and regulations evolve (Pourmohammad-Zia & van Koningsveld, 2024). Functionally, the digital twin is a unifying platform that integrates multiple models within a single system and is equipped with advanced analytical capabilities. Owing to its coherent architecture and online accessibility, the digital twin serves as a real-time decision-support system for last-mile stakeholders, enabling

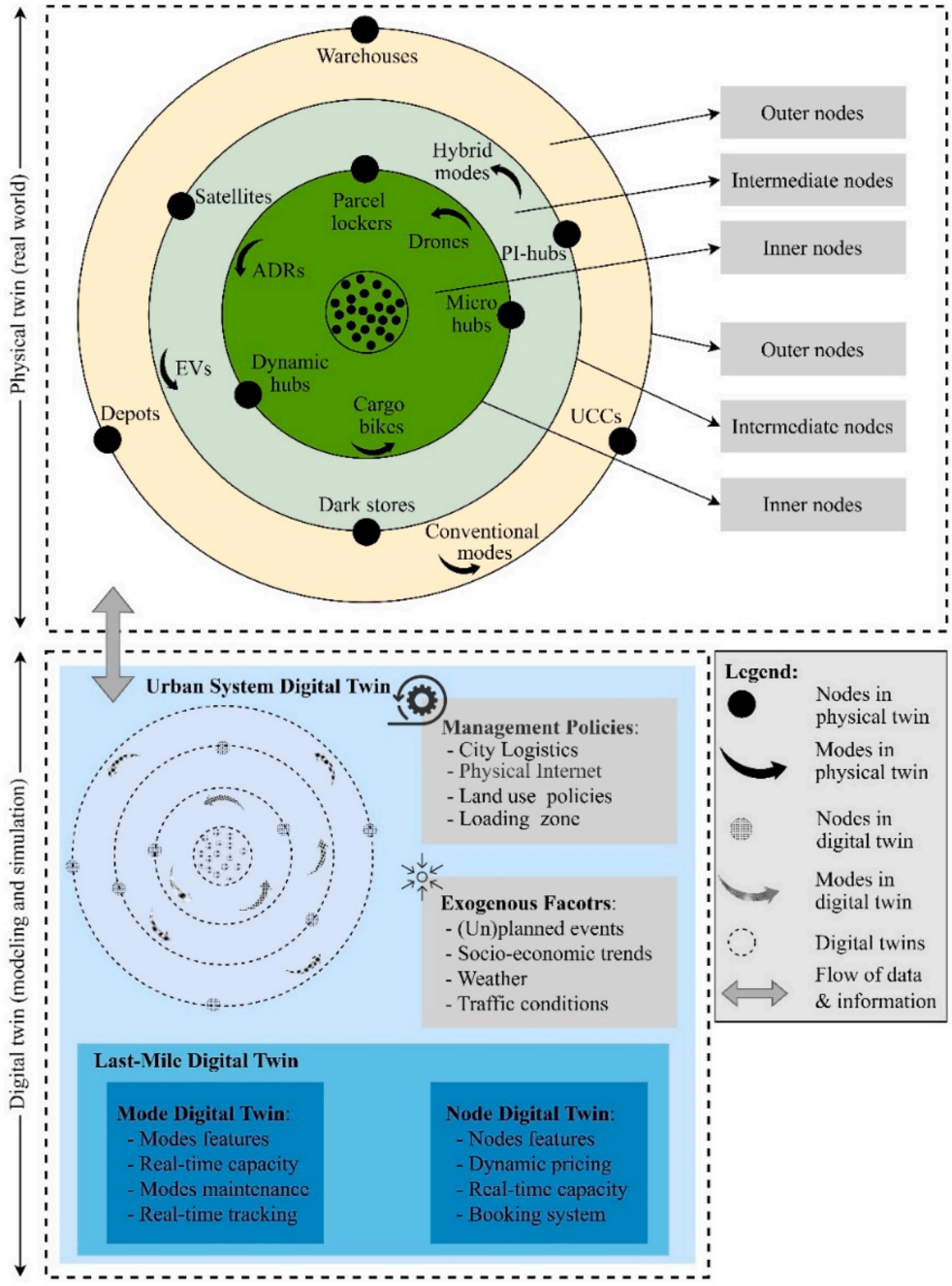


Fig. 6. Overview of the transportation nodes, modes, and network structure embedded in the physical twin and digital twin of the LM-LRP.

simulation, forecasting, and optimization analyses under dynamic and uncertain operating conditions. Thus, in the LM-LRP context, the digital twin is not considered here as a new mathematical solution method in itself, but as an enabling paradigm that can support more responsive decision-making by connecting real-time system states with simulation, forecasting, and optimization models.

Presenting the digital twin alongside its physical twin makes its integrative role more apparent. Accordingly, we have divided the entire LM-LRP system into two parts, the physical twin and the digital twin as shown in Fig. 6. This figure illustrates how LM-LRP components in the real world connect with one another and with the twin part. The physical twin represents the structural components of the LM-LRP and captures the coupling between nodes and modes across the distribution network. The framework is organized as a radar-style diagram, in which the spatial ordering naturally defines three conceptual echelons, namely an inner echelon, one or more intermediate echelons, and an outer echelon. Customers are located at the center, and different nodes are positioned outward based on their proximity to customers and their functional roles. Similarly, appropriate modes are inherently interdependent with the echelon structure. For example, higher-capacity and longer-range modes, e.g., medium trucks and vans, typically connect the outer and intermediate echelons, whereas lower-capacity and more agile modes, e.g., cargo bikes and ADRs, are suitable for the inner echelon.

The digital twin has two primary functions, to mimic the physical twin through simulation and optimization models, and to provide analytics that inform decision-makers. First, the digital twin must be a reasonably accurate representation of the physical twin, which depends on the quality of the modeling. We divide these models into the mode digital twin and the node digital twin, which are representative of the main LM-LRP components. A mode twin tracks and records delivery-vehicle operating states, including real-time location and speed, state of charge for electric fleets or fuel level for conventional fleets, parcels carried and delivered, and maintenance condition. A node twin captures capacity, location, and the number of logistics hubs, in addition to real-time availability, scheduling, and service fees. Since the last mile occurs in urban areas, an urban digital twin is needed to represent higher-level urban conditions and maximize modeling accuracy in the digital twin. These conditions can include policies on parking and loading zones, real-time traffic conditions, and planned or unplanned road closures.

Apart from the simulation and optimization aspects needed to mimic the physical twin, the digital twin also enforces safety and security protocols. For example, by employing PI-compatible identifiers, traceability across handovers between logistics nodes is enhanced and a higher level of security is reinforced. Finally, a well-developed digital twin is equipped with various analytics that generate insights from real-time and historical data. These results, combined with predictive analyses, are fed back to the physical twin, forming a continuous information flow loop that enables data-driven decision-making in the LM-LRP.

A critical requirement for establishing a digital twin in the LM-LRP is the availability of real-time data, which can be grouped into three main categories: traffic data, global positioning system (GPS) data, and Internet of Things (IoT) data. Traffic data are required when formulating the LM-LRP, as congestion directly affects travel time and thus influences routing decisions and feasible hub locations, particularly in highly populated areas where variability in traffic conditions reduces system efficiency (Du et al., 2023; Guo et al., 2024; Rautela et al., 2022). In addition, GPS data enable continuous tracking of vehicles and goods, supporting real-time location awareness and improving routing responsiveness in last-mile operations (Çoban et al., 2021; Singh et al., 2018). IoT further extends this capability by allowing interconnected physical entities to autonomously sense and acquire high-resolution environmental and operational data through sensor networks (Xu et al., 2025). This continuous data exchange supports integrated monitoring and synchronization, and enables coordinated decision-making across the physical and digital twins (Thompson et al., 2025).

4. Distribution network structure

Network structure, defined by the number of echelons, is critical in LM-LRP design because it governs parcel flows from origin to destination. Network structure also shapes the feasibility of employing emerging initiatives. Finally, some LM-LRP applications, e.g., pickup and delivery, are more readily adopted in certain network structures. Hence, in this section, we classify LM-LRP structures and then discuss which emerging initiatives and applications are most suitable for each structure.

4.1. Single-echelon – location routing problem

A single-echelon LRP (1E-LRP) represents a distribution system with a single delivery layer. Operating nodes are selected based on capacity requirements and proximity to customers, and goods are transported directly from selected nodes to customers. Core decisions include node opening, customer allocation, mode selection and deployment, and route design. For mode selection, characteristics such as capacity, speed, fuel type, energy consumption, emissions, and capital costs are usually considered.

In the 1E-LRP, objectives have evolved from cost-centric formulations, based on node opening and mode capital and operational costs, to multidimensional performance measures that include emissions, profit, and service quality (Ahmadi-Javid et al., 2018; Bruni et al., 2023; Koç, 2016; Liu et al., 2023; Schneider & Löffler, 2017; Su et al., 2025). In the 1E-LRP literature, technological innovations are mainly related to transportation modes. EVs are the most incorporated mode initiative, employed to satisfy various tasks from last-mile delivery to waste collection (Kahalimoghadam & Thompson, 2026; Mirzavand Boroujeni et al., 2025; Schiffer & Walther, 2018; Yang et al., 2022). Pickup and delivery, an LM-LRP variant in which customers' demands are picked up and delivered within the same route, is the main functional extension that fits well within the LM-LRP when EVs are used (Leng et al., 2023). Drones are another initiative used for small-package delivery (Bruni et al., 2023). However, incorporating drones into the model often requires a stochastic formulation, as drone flight time can be affected by wind speed and weather conditions and is uncertain. ADRs have also been adopted as a mode initiative in the 1E-LRP to fulfill e-commerce demand, mainly for groceries and medicine (Ghiani et al., 2025). The literature suggests that ADR deployment can improve sustainability, as they are battery-powered and are commonly categorized as net-zero-emission modes. In addition, cargo bikes are another sustainable mode specifically used for parcel delivery (Rave & Fontaine, 2025).

On the node side, PLs have been introduced into the network for last-mile delivery, particularly for priority e-commerce deliveries (Schiffer et al., 2018). Another application reported in the literature is mobile PLs, where PL locations are dynamically relocated based on customer locations (Mirzavand Boroujeni et al., 2025). Compared with home delivery, which is associated with higher failed-delivery rates and greater vehicle-kilometers traveled, PLs improve sustainability by enabling customers to collect parcels from these nodes, thereby reducing re-delivery trips. Micro-hubs are also employed in the 1E-LRP as release points for cargo bikes, in addition to providing storage and consolidation functions.

Table 4 provides a structured summary of the 1E-LRP literature, covering objective-function form (single or multi-objective), embedded technologies in the network, main application, and solution methodology (exact, heuristic, or metaheuristic). The table also flags whether a study considers sustainability, stochastic behavior, pickup and delivery operations, or synchronization, which are the key functional extensions reported in the literature.

Table 4
Overview of 1E-LRP.

Reference	Objective function		Technology	Application	Methodology	Variety
	Type	Components				
Koç (2016)	Single	Economic: Facility activation; Vehicle fixed; Vehicle operational	NA	Periodic service with time windows	Heuristic: Adaptive large neighborhood search (ALNS); mixed-integer linear programming (MILP) formulation	NA
Koç et al. (2016)	Single	Economic: Facility activation; Vehicle fixed & operational; CO_2 emissions	NA	Urban freight (generic)	Heuristic: ALNS	Sustainability
Singh et al. (2018)	Single	Economic: Facility activation; Vehicle operational; Refrigeration & storage energy;	NA	Cold chain; travel distance capped by shelf-life	Exact: Formulated as MILP solved using IBM ILOG CPLEX 12.5[a]	NA
Schneider & Löffler (2017)	Single	Economic: Facility activation; Vehicle fixed; Vehicle operational	NA	Urban/city logistics (generic)	Heuristic: Branch-and-Bound + Granular Tabu Search	NA
Ahmadi-Javid et al. (2018)	Single	Economic: Revenue; Purchasing; Facility activation; Vehicle operational	NA	Retail/distribution (generic)	Exact: Branch-and-Price / Dantzig-Wolfe decomposition.	NA
Liu et al. (2023)	Single	Economic: fleet size, travel distance, service-delay penalties	NA	Last-mile parcel delivery	Heuristic + Learning: Hybrid Q-Learning with global and local search;	NA
Schiffer et al. (2018)	Single	Economic: Facility activation (intra-route); Vehicle fixed; Vehicle operational	EVs	Last-mile distribution	Heuristic: ALNS for routing + facility-configuration lower bound	Sustainability
Schiffer & Walther (2018)	Single	Economic: Facility activation; Vehicle fixed & operational	EV charging	City logistics; waste collection	Heuristic: ALNS + dynamic programming/local search	NA
Yang et al. (2022)	Single	Economic: Vehicle fixed; Vehicle operational	EVs	Last-mile logistics	Matheuristic: Alternating direction method of multipliers (ADMM)	Pickup–delivery with time windows
Bruni et al. (2023)	Single	Social: Latency (sequence-weighted waiting);	Drones	Small-package last-mile	Exact: Branch-and-Check with custom no-good cuts	Stochastic; Uncertainty
Leng et al. (2023)	Single	Economic: Facility activation; Vehicle fixed & operational; CO_2 cost	EVs	Urban parcel/ freight	Heuristic: Q-learning hyper-heuristic within simulated annealing	Sustainability; Pickup and delivery
Ghiani et al. (2025)	Single	Economic: Facility activation (public transport stations for ADRs); Vehicle operational	ADRs	E-commerce, grocery, medical supplies	Matheuristic: location-based set-partitioning and heuristic,	Sustainability
Mirzavand Boroujeni et al. (2025)	Single	Economic: Facility activation, PLs; Vehicle fixed & operational; Prize maximization	EVs; PLs;	High-priority e-commerce	Heuristic LNS with simulated annealing	NA
Rave & Fontaine (2025)	Single	Economic: Facility activation; Vehicle fixed & operational	Cargo bikes; micro-hubs;	Last-mile parcel delivery	Heuristic: ALNS with tactical allocation split; compact MILP baseline	NA
Su et al. (2025)	Multi	Economic: Facility activation; Vehicle fixed & operational, Environmental: Carbon treatment; Social: Service deviation penalties	NA	Urban e-commerce express	Metaheuristic: Non-dominated sorting genetic algorithm II (NSGA-II); Pareto front	Sustainability
Wang et al. (2025)	Single	Economic + Environmental: minimize total operating cost (facility + vehicle + distance + weight-based CO_2 emissions)	NA	Urban logistics	Exact: Branch-Price-and-Cut algorithm with bidirectional labeling, rounded-capacity	Sustainability

[a] An exact commercial solver employing branch-and-bound and branch-and-cut algorithms.

4.2. Two-echelon network

4.2.1. Two-echelon location routing problem

In the two-echelon LRP (2E-LRP), there is an additional node layer between the origin and destination. This additional layer increases the flexibility of the 2E-LRP compared with the 1E-LRP, as the additional nodes are used for storage, consolidation, or transshipment. Although mode initiatives in the 2E-LRP overlap with those in the 1E-LRP, recent studies show that the applications of these modes have expanded, including truck-drone delivery (Kitjacharoenchai et al., 2020) and integrated drone-and-ADR operations (Liu et al., 2021). More notably, in recent years, node-related initiatives have received greater attention. For example, satellites are used as dynamic facilities that support drones and ADRs in e-commerce fulfillment (Liu et al., 2021) or as temporary facilities deployed to respond to demand fluctuations (Sutrisno & Yang, 2023). PLs' roles have also expanded from solely acting as collection nodes (Amini & Haughton, 2023; Liu et al., 2021; Ozyavas et al., 2025; Zhao et al., 2025) to serving as autonomous (Li et al., 2021a) or mobile nodes (Korkmaz et al., 2025) to accommodate the rise in delivery frequency. In all of these cases, system complexity increases, as the modeling includes synchronization between first-echelon arrivals and second-echelon departures (Agnimo et al., 2023; Sutrisno & Yang, 2023), as well as heterogeneous modes (Kitjacharoenchai et al., 2020; Zhao et al., 2018).

The 2E-LRP covers a wide variety of last-mile applications. Urban last-mile delivery is a primary application area, with many studies aiming to enhance sustainability, mainly by reducing CO_2 emissions (Du et al., 2023; Simoni et al., 2018) or by enabling collaborative networks (Aloui et al., 2021; Du et al., 2023; Ouhader & El Kyal, 2023). Addressing urban challenges, such as high delivery volumes in dense areas (Agnimo et al., 2023) or delivery in congested areas (Escobar-Vargas & Crainic, 2024) is another common application, often enabled by incorporating synchronization into the model. Implementing pickup and delivery logistics systems is another application enabled by the 2E-LRP structure. In this application, network efficiency and sustainability are improved by combining environmentally friendly mode initiatives, e.g., cargo bikes or EVs, with autonomous nodes, e.g., PLs (Li et al., 2021a; Ozyavas et al., 2025; Wang et al., 2022). Lastly, uncertainty, mainly in demand, is included in the 2E-LRP to make simulations and models more realistic (Amini & Haughton, 2023; Guo et al., 2024; Zhao et al., 2025). Although this improves system accuracy, it increases modeling complexity, as it requires stochastic programming to cope

with uncertainty. Table 5 synthesizes objectives, technologies, applications, and methods across these studies.

4.2.2. Two-echelon vehicle routing problem

Under the 2E network, the two-echelon vehicle routing problem (2E-VRP), an extension of the VRP focused on routing optimization, is also included in our review, as some studies incorporate node decisions alongside routing. In this stream, the routing component of the LM-LRP is more prominent, with applications emphasizing mode- and service-specific features such as occasional drivers (Yu et al., 2021), multi-commodity flows (Dahimi et al., 2025), and multi-temperature deliveries (Xu et al., 2025). From the nodes and modes perspective, PLs (Enthoven et al., 2020; Yu et al., 2021), EVs (Anderluh et al., 2021), and ADRs (Liu et al., 2022; Yu et al., 2022) are the most common initiatives in the literature. While synchronization is the dominant functional extension identified in 2E-VRP, sustainability and pickup and delivery are the other evaluated themes. Table 6 summarizes the main components of studies conducted in 2E-VRP.

4.3. Multi-echelon LRP

Multi-echelon LM-LRP refers to a network that, in addition to the inner and outer echelons, has more than one intermediate echelon. It extends the 2E-LRP concept by dividing the network into additional layers, thereby enhancing flexibility and efficiency. Such a network captures upstream goods flows from higher levels, typically starting with suppliers, then plants and manufacturers, followed by local transit (consolidation) nodes, and finally customer distribution centers (Dai et al., 2019). Key advantages of multi-echelon systems include lower last-mile cost, improved node-capacity management, and greater resiliency. Other advantages of multi-echelon LRP include the use of dynamic hubs to handle time-varying demand and reduce operating costs (Kır & Comert, 2024), as well as improved efficiency through optimal task allocation based on distance and load (Wang et al., 2024). However, these designs may require higher opening costs, as multiple layers of nodes are needed (Dai et al., 2019). Table 7 summarizes multi-echelon LRP papers.

4.4. Adaptive echelon topologies

Dynamic echelon networks allow intermediate facilities to change roles over time in response to demand. Such systems are suitable for omnichannel manufacturers seeking to integrate depots, dark stores, retail outlets, and customer zones (Tahirov et al., 2025). Network flexibility is high, as the structure can be modified based on business conditions, e.g., demand and targeted profit margins (Kır & Comert, 2024). Node roles can be reassigned across defined planning periods, enabling re-optimization of node locations and mode allocation over time. Finally, adaptive echelon LRP is well aligned with the PI concept, which promotes flexible distribution network design (Kahalimoghadam et al., 2025b).

5. Current operational landscape of the LM-LRP

This section synthesizes the current operational patterns in the reviewed studies, focusing on how node and mode combinations shape performance and trade-offs in the LM-LRP systems.

5.1. Operational patterns and trade-offs in the LM-LRP

Centralized depots and UCCs provide high consolidation and simpler fleet control, but they may increase the distance to customers and intensify congestion, parking pressure, and emissions in dense areas. In contrast, micro-hubs and satellites improve customer proximity and support walking, cycling, cargo-bike, and robotic delivery, but they require additional replenishment, storage, and synchronization. Similarly, PLs and pickup stations reduce failed deliveries and consolidate demand, but their effectiveness depends on customer acceptance, accessibility, and locker-capacity management. Mobile hubs and mobile PLs provide greater spatial flexibility, especially under variable demand, but they increase routing interdependence because the location and timing of the mobile node must be coordinated with downstream delivery modes.

ADRs and drones improve delivery flexibility and reduce labor dependency, but remain constrained by payload capacity, operating range, battery limitations, safety concerns, infrastructure suitability, weather conditions, and regulatory acceptance. This trade-off between automation potential and deployment feasibility makes ADRs and drones more suitable as complementary modes within specific operating environments rather than universal replacements for conventional delivery.

Apart from nodes and modes, how delivery is operated is another dimension discussed in the literature. Crowdshipping and collaborative delivery use available capacity from individuals, carriers, or shared networks and can improve asset utilization and provide flexible capacity, particularly during peak-demand periods. However, they create challenges related to reliability, incentive design, privacy, and service quality. Unlike dedicated fleet systems, crowd-based capacity may not be available when and where it is needed.

Overall, the reviewed studies show that LM-LRP research has moved from centralized, vehicle-based distribution toward more decentralized, multimodal, and collaborative configurations. This shift is driven by urban congestion, emission targets, demand variability, customer-service expectations, and limited access to urban logistics space.

5.2. Cross-cutting implications for modeling the LM-LRP

These trade-offs become more complex when node and mode decisions are considered together. Improving proximity through micro-hubs, satellites, or mobile hubs may reduce final delivery distance, but it increases the number of nodes that must be supplied and coordinated. Similarly, using smaller or cleaner modes can improve accessibility and environmental performance in dense areas, but these modes often have lower payload capacity, shorter operating range, and stronger dependence on local infrastructure. Mobile and multimodal systems add flexibility under variable demand, but this flexibility comes with stronger synchronization requirements between facilities, vehicles, couriers, and delivery schedules.

The reviewed studies also suggest that many initiatives become more operationally meaningful when combined with complementary components. For example, low-emission modes are more effective when supported by nearby micro-hubs, satellites, or UCCs, as shorter travel distances help overcome their limited range and capacity. Similarly, ADRs and drones are more suitable when operating from distributed or mobile nodes rather than from distant centralized facilities. PLs and pickup stations can improve delivery consolidation, but their benefits depend on their integration with routing decisions, customer-access patterns, and capacity management.

A further pattern is that combining initiatives shifts the main challenge from routing efficiency to system coordination. As more nodes, modes, and actors are introduced, the problem becomes less about finding the shortest or cheapest route and more about coordinating replenishment, transshipment, timing, capacity, customer collection behavior, and stakeholder participation. This is particularly evident in multimodal, mobile, autonomous, and collaborative systems, where poor synchronization can reduce or eliminate expected benefits.

Table 8 summarizes the main node–mode combinations identified across the reviewed studies and highlights their associated advantages, limitations, and core operational trade-offs.

Additionally, Fig. 7 illustrates the interdependencies between the major LM-LRP objectives and the identified technology-enabled initiatives. In this diagram, the main objectives are shown in blue, performance targets are shown in yellow, and operational enablers are shown

Table 5
Overview of 2E-LRP.

Reference	Objective function		Technology	Application	Methodology	Variety
	Type	Components				
Nguyen et al. (2012)	Single	Economic: Facility, vehicles, and routing costs	NA	Urban last-mile	Heuristic and metaheuristic: greedy randomized combined by Clarke and Wright	NA
Simoni et al. (2018)	Multi	Economic: cost; Environmental: CO_{22}	NA	Urban last-mile	Metaheuristic: Genetic Algorithm and Memetic Algorithm	Sustainability
Gianessi et al. (2016)	Single	Economic: routing; facility.		First-echelon ring design	Hybrid; Exact: CPLEX Matheuristic: nearest neighbor	NA
Janjevic et al. (2019)	Single	Economic: satellite fixed & variable; Vehicle operating;	NA	Urban last-mile e-commerce	Hybrid; Exact: Gurobi; Heuristic: greedy/constructive heuristics and reduced enumeration	NA
Agnimo et al. (2023)	Single	Economic: facility, routing, costs	NA	Dense urban transfers	Exact: Binary variable fixing method solved with Gurobi	Synchronization
Escobar-Vargas and Crainic (2024)	Single	Economic: Facility opening and routing costs	NA	Congested urban transfers	Exact: dynamic discretization discovery	Synchronization
Ozyavas et al. (2025)	Single	Economic: costs of opening PLs and routes	PL; cargo bike	Urban delivery with flexible hubs	Exact: branch-and-price	Pickup and delivery
Du et al. (2023)	Multi	Economic: vehicles and satellite costs; Environmental: CO_2; Social: Service satisfaction	EVs	Collaborative urban last-mile	Heuristic: hybrid immune algorithm + improved NSGA-II	Sustainability
Chen et al. (2024b)	Single	Economic: vehicle, cooling, labor and time penalty costs	NA	Mixed-mode last-mile	Heuristic: ALNS	Pickup and delivery
Kitjacharoenchai et al. (2020)	Single	Economic: Total truck arrival time	Drones (truck–drone)	Parcel delivery with drone assist	Hybrid – MIP (GAMS/CPLEX) for small instances + LNS for larger instances	Synchronization
Liu et al. (2021)	Multi	Economic: transportation, handling, satellite opening costs; Environmental: emissions	Drones; ADRs; dynamic satellites; PLs	Urban e-commerce last-mile	Heuristic: Artificial immune algorithm and NSGA-II	Synchronization
Korkmaz et al. (2025)	Single	Economic: total operational cost	Mobile PLs; EV trucks	Repositionable PL networks	Hybrid – Exact: integer linear programming (IBM OPL / CPLEX); Heuristic: Two-stage decomposition	Synchronization;
Sutrisno and Yang (2023)	Single	Economic: facility, vehicles, and routing costs	Mobile and temporary satellites	Demand-responsive repositioning	Heuristic: Clustering-based simultaneous neighborhood search	Synchronization
Tian and Hu (2023)	Single	Economic: total travel cost	Drones	Pop-up satellites in dense areas	Hybrid; Exact: column-generation and Branch-and-Price. Heuristic: Sequential route extraction	NA
Zhou et al. (2017)	Single	Economic: total delivery cost	NA	E-commerce; Home delivery & pickup by customers	Heuristic: genetic algorithm + simulated annealing	Stochastic
Ouhader and El Kyal (2023)	Multi	Economic: facility opening, transshipment, vehicles fixed & operational; Environmental: CO_2	NA	Multi-period collaborative last-mile	Exact: MILP solved via commercial solver (Branch and Bound in MATLAB/CPLEX) with ε-constraint for Pareto sampling	Sustainability; Collaborative
Aloui et al. (2021)	Multi	Economic: fixed facility, vehicle, inventory, and transport costs Environmental: load-dependent + distribution center operation emissions. Social: accident rate	NA	Multi-period two-echelon collaborative inventory-location-routing	Hybrid – Exact: MILP solved by CPLEX for small instances Heuristic: modified k-means clustering + genetic algorithm for large instances	Sustainability; Collaborative
Ahmadi et al. (2015)	Single	Economic: travel time, penalty for unsatisfied demand, and fixed depot opening costs	NA	Humanitarian (earthquake logistics)	Hybrid – Exact: CPLEX for small instances; Metaheuristic: Variable neighborhood search (VNS) algorithm	Stochastic; Uncertainty
Amini and Haughton (2023)	Single	Economic: depot opening cost, expected routing cost and waiting cost	PLs	E-commerce with availability scenarios	Exact: scenario-based MILP	Stochastic
Guo et al. (2024)	Single	Economic: worse-case depot lease cost and routing costs	NA	Urban freight (leasing volatility)	Hybrid – Exact: robust-counterpart transformation. Heuristic: Iterative Greedy; ALNS	Uncertainty
Li et al. (2021a)	Single	Economic: routing, depot rental, and rendezvous points costs	Autonomous mobile PLs	Urban PL networks	Heuristic: Hybrid Clarke & Wright	Pickup and delivery; Synchronization; Sustainability
Yu et al. (2020)	Single	Economic: satellite set-up, vehicle activation & routing, and loading/unloading	NA	Express delivery of grocery/hypermarket products	Hybrid; Exact: CPLEX. Heuristic: Simulated annealing	NA
Zhao et al. (2018)	Single	Economic: intermediate depot and routing costs	NA	Joint parcel delivery	Hybrid; Exact: Gurobi; Heuristic: Hybrid Clarke & Wright variants	NA

(*continued on next page*)

Table 5 (*continued*)

Reference	Objective function		Technology	Application	Methodology	Variety
	Type	Components				
Chen et al. (2024a)	Single	Economic: Priority-related time costs	Drone	Multi-fleet last-mile delivery	Hybrid; Exact: Gurobi; Heuristic: multi-start improved ALNS + ant colony	NA
Rautela et al. (2022)	Single	Economic: Fixed & variable operating costs of satellites and routing cost	NA	Urban last-mile e-commerce	Exact: Gurobi	Pickup and delivery
Wang et al. (2022)	Single	Economic: opening and handling costs of pickup stations; vehicles fixed & routing.	EVs	Urban last-mile distribution with pickup stations	Exact: branch-and-price with column generation	Pickup and delivery
Wang et al. (2023)	Multi	Economic: transportation, pickup and delivery, maintenance, and facility fixed costs; time windows penalties; Fleet size	NA	City logistics	Metaheuristic: 3D k-means clustering for initialization; hybrid particle swarm optimization with Clarke and Wright savings and variable neighborhood search.	Pickup and delivery
Juvvala and Sarmah (2025)	Single	Economic: Energy consumption, UCC opening, carbon-emission costs, and restricted-zone entry fees	EVs	Urban grocery distribution	Heuristic: Particle Swarm Optimization with Differential Evolution	Sustainability; Stochastic
Pourmohammad-Zia and van Koningsveld (2024)	Single	Economic: travel costs and facility establishment costs	Light EVs	Urban multimodal distribution	Hybrid solution: ALNS for second-echelon location and routing; branch-and-price for first-echelon routing	Synchronization
Zhao et al. (2025)	Single	Economic: pickup-stations opening costs, service costs, delivery costs.	PLs; self-collection points	Green last-mile parcel delivery	Metaheuristic: two-stage hybrid simulated annealing – Attention Model	Stochastic

Table 6
Overview of 2E-VRP.

Reference	Objective function		Technology	Applications	Methodology	Variety
	Type	Components				
Breunig et al. (2016)	Single	Economic: routing, handling, fixed vehicle, facility opening	NA	City-logistics goods delivery	Heuristic: LNS	Synchronization
Hemmelmayr et al. (2012)	Single	Economic: routing/operating	NA	City-logistics goods delivery	Heuristic: ALNS	NA
Enthoven et al. (2020)	Single	Economic: PL and routing costs	PLs	City-logistics / e-commerce deliveries	Hybrid – Exact: CPLEX; Heuristic: ALNS	NA
Anderluh et al. (2021)	Multi	Economic: vehicle routing & fixed costs Environmental: greenhouse gas emissions; Social: noise & congestion disturbance	EVs	Urban city-logistics deliveries	Heuristic: ALNS	Synchronization; Sustainability
Yu et al. (2021)	Single	Economic: routing and customers covering costs	PLs	Last-mile delivery: home delivery or pickup	Heuristic: ALNS	Synchronization
Liu et al. (2022)	Single	Economic: depot & micro-hub opening, routing, load-dependent and emission-dependent costs	ADRs	E-grocery last-mile delivery	Hybrid – Exact: Gurobi. Heuristic: clustering-based artificial immune algorithm	Synchronization
Yu et al. (2022)	Single	Economic: routing costs for vans and robots	ADRs	Hybrid urban pickup and delivery	Heuristic: ALNS	Synchronization
Zhou et al. (2024)	Single	Economic: routing and vehicles costs	NA	City-logistics last-mile delivery	Hybrid – Exact: CPLEX; Heuristic: ALNS	Synchronization
Xu et al. (2025)	Multi	Economic: vehicle fixed & operational, refrigeration, and damage costs; Environmental: CO_2 costs; Social: freshness and delivery satisfaction	NA	Community group-buying logistics	Heuristic: Adaptive Genetic Hyper-heuristic with multiple local search operators	Sustainability

in white. The identified technologies and initiatives that facilitate the achievement of these operational enablers are grouped in bordered boxes.

6. Future of the LM-LRP

This section develops an integrated framework to guide the incorporation of emerging technologies into the LM-LRP and address existing research gaps, serving as a blueprint for future studies.

6.1. Identified gaps

6.1.1. Limited integration of emerging technologies into the LM-LRP

The literature has largely evaluated the impacts of node initiatives and mode initiatives separately. However, many of these initiatives have a higher implementation likelihood when they operate alongside one another. A tangible example is the integration of PLs, PI-hubs, and micro-hubs, which can be located close to customers, with ADRs and drones to automate the final leg of delivery and increase the utilization of these nodes. In turn, the entire distribution network and its stakeholders can benefit from these advantages, including improved

Table 7
Overview of Multi-echelon LRP.

	Objective function					
Reference	Type	Components	Technology	Applications	Methodology	Variety
Dai et al. (2019)	Single	Economic: Facility opening, vehicles fixed, & routing	NA	Urban supply-chain distribution	Heuristic: improved Clarke and Wright savings algorithm	Stochastic
Arevalo-Ascanio et al. (2024)	Single	Economic: Facility opening, vehicles fixed, & routing costs	NA	Strategic model for last mile	Method: continuous multi-scale approximation and districting	Stochastic; Uncertainty
Kır and Comert (2024)	Single	Economic: Minimize total travel distance	NA	Last-mile e-commerce	Heuristic: Ant colony optimization	Dynamic hub
Wang et al. (2024)	Multi	Economic: Transport & facility costs Environment: carbon emissions	EVs	Green last-mile logistics	Heuristic: Optics temporal-spatial discretization combined with multi-strategy slime mould algorithm.	Sustainability

Table 8
Summary of node-mode combinations and core trade-offs in LM-LRP.

Node/mode combination	Main advantage	Main limitation	Core trade-off
Depot/UCC + vans	High capacity and simple control	Congestion, emissions, distance from customers	Consolidation versus proximity
UCC + low-emission vehicles	Reduced urban externalities	Extra handling and facility costs	Sustainability versus economic viability
Micro-hub/ satellite + bikes/walking	High proximity and local accessibility	Limited capacity and replenishment needs	Accessibility versus operational stability
PLs/pickup points	Fewer failed deliveries and better consolidation	Customer inconvenience and locker-capacity limits	Carrier efficiency versus customer accessibility
Mobile depot/ mobile locker	Flexible response to spatial demand	Strong timing and routing interdependence	Flexibility versus synchronization complexity
Two-echelon multimodal systems	Functional specialization of vehicles and modes	Transshipment and coordination burden	Efficiency gains versus synchronization costs
ADRs/drones	Labor-saving and flexible delivery	Payload, range, safety, and regulatory constraints	Automation potential versus deployment feasibility
Crowdshipping/ collaborative modes	Asset-light capacity and scalability	Reliability, incentives, and trust issues	Flexibility versus control
Shared/PI-inspired hubs	System-wide resource utilization	Governance, data sharing, and benefit allocation	Network efficiency versus institutional complexity

suitability for agile delivery and reductions in total vehicle-kilometers traveled and associated emissions. Finally, there is a need for new benchmarks tailored to technology-enabled LM-LRPs, such as van–drone or van–ADR deliveries in networks that include PI-hubs and micro-hubs, to enable systematic solution comparison and foster more rigorous, innovative methodologies.

6.1.2. *Inadequate representation of the real world*

While many real-world applications have been reflected in LM-LRPs, existing models do not fully capture real-world complexity. Stochastic variability arising from uncertain travel times and unpredictable customer behavior, combined with the NP-hard nature of the LM-LRP, often leads to simplifying assumptions. The former issue has been examined through demand modeling, which reports that demand can be a function of temporal factors such as time of year, promotions, product category, historical sales patterns, and customer and market behavior (Li et al., 2021b; Salamai et al., 2022). However, the literature lacks integrated frameworks that combine demand forecasting models with LM-LRPs to produce more feasible and practical solutions. The response to the latter issue has mainly focused on developing (meta)heuristic algorithms. While these approaches significantly reduce computation time compared with exact methods, they cannot guarantee global optimality.

6.1.3. *Governance and regulatory gaps*

The rise of open and shared nodes necessitates new regulatory mechanisms to manage and coordinate them. Only a limited number of studies have incorporated this dimension into logistics modeling. These studies mainly introduce a new stakeholder, commonly referred to as the PI-manager or PI-orchestrator, who is responsible for coordinating shared facilities and harmonizing vertical and horizontal collaborations. While this provides a useful starting point, further investigation is required to examine the role of this stakeholder across different geographical contexts and to evaluate the feasibility of adopting PI-hubs under local regulations and their acceptability among other stakeholders. Additional policy changes by government authorities are also required for the deployment of high-tech transportation modes. While ADRs and drone deliveries have already been adopted in some countries, many governments remain conservative and express concerns regarding security and public safety. Alongside technical assessments demonstrating the economic and environmental viability of these technologies, workshops and conferences can help improve acceptability by increasing awareness and supporting knowledge sharing.

6.1.4. *Technology-specific gaps and conflicting findings*

Beyond these broader gaps, the reviewed studies also reveal technology-specific limitations, contradictions, and underdeveloped findings, as summarized in Table 9. The table also presents sustainability trade-offs between economic, social, and environmental perspectives. These issues are important because the effects of emerging technologies on LM-LRP performance vary depending on the modeled objective, deployment context, supporting infrastructure, and operational assumptions.

6.2. *Future research directions*

This section aims to establish a comprehensive framework based on the identified gaps. We built this framework in alignment with the widely accepted PI roadmap developed by the Alliance for Logistics Innovation through Collaboration in Europe (ALICE), a comprehensive blueprint for transitioning to a zero-emission and green logistics ecosystem by 2040 (Ballot et al., 2020). The roadmap is structured around five areas of development: Logistics Nodes, Logistics Networks, System of Logistics Networks, Access and Adoption, and Governance. Accordingly, an integrated framework is developed, as shown in Fig. 8, to outline clear actions required to advance the LM-LRP. Each area of development is divided into five generational phases, whereby the first two generations represent the review findings. More importantly, the last three generations specify actions required to address the existing gaps in the LM-LRP. The remainder of this section discusses how these actions can be translated into future research directions and practical applications.

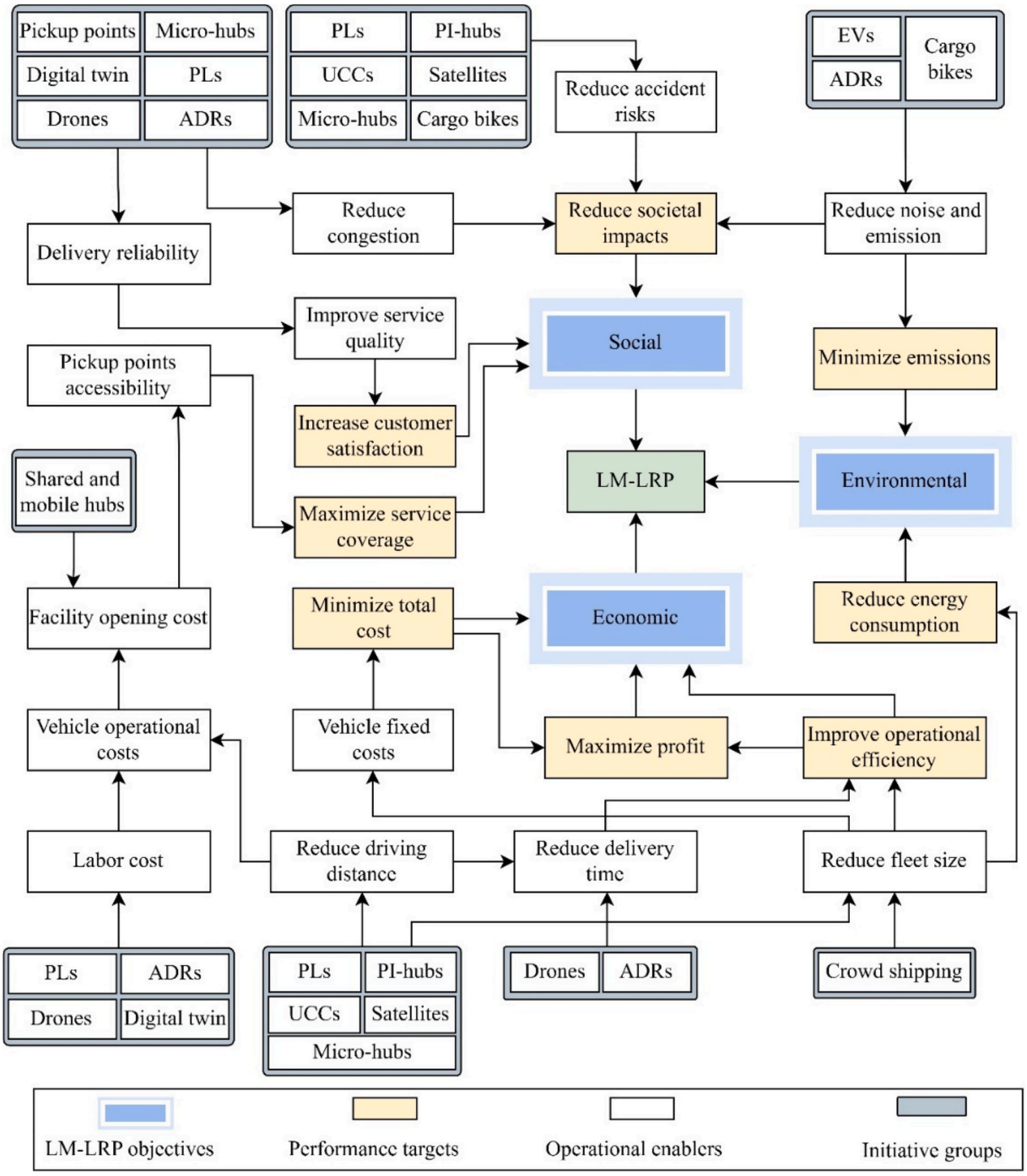


Fig. 7. Interdependencies between technology-enabled initiatives and major objectives in the LM-LRP.

6.2.1. Logistics nodes

Generation 1: The shift from traditional depots and warehouses toward multifunctional hubs, especially UCCs, began. While this shift increased consolidation levels, it often lacked economic sustainability and relied on public support. This was because, first, acceptance of the consolidation concept was low, and second, UCCs were relatively large nodes that required sufficient demand volume. To overcome these limitations, the next generation emphasizes decentralization and scalability.

Generation 2: The limitations are addressed by adopting smaller and more distributed facilities, including micro-hubs and dark stores. Their reduced scale lowers the demand threshold needed for viability, making the nodes economically feasible. They are also well suited to the growing needs of e-commerce, including rapid delivery and narrow time windows. Automated pickup nodes, particularly PLs that provide 24/7 customer access, also expanded in this period. However, realizing their full potential requires integration with automated modes to improve operational efficiency.

Generation 3: The distribution network design will move toward integrating automated nodes and modes. In this phase, autonomous technologies such as ADRs and drones increasingly undertake the last leg of delivery, improving operational flexibility and efficiency. Distribution networks that locate PLs while considering the specifications of ADRs and drones are expected to become more prominent. With increasing reliance on automated operations, the need for coordination across distributed assets becomes more critical, leading to system-level

Table 9
Technology-specific gaps and sustainability trade-offs for each initiative in the LM-LRP.

Initiative/ Sustainability	Identified gaps in the literature
ADRs	Cost benefits are not universal. ADR systems can reduce costs in some settings, but their competitiveness depends on scenario conditions, fixed costs, robot travel cost, and supporting infrastructure (Ghiani et al., 2025; Liu et al., 2022; Yu et al., 2022). Emission results depend on modeling assumptions. Some studies treat ADRs as near zero-emission vehicles, while others use load-dependent emission formulations (Ghiani et al., 2025; Liu et al., 2022). Access–feasibility trade-off remains critical. ADRs can improve access to dense or van-restricted areas, but their feasibility is limited by range, capacity, speed, and autonomy (Ghiani et al., 2025; Liu et al., 2022; Yu et al., 2022).
Sustainability trade-off*	ECN: ↑↓ ENV: ↑↓ SOC: ↑ ADRs improve social sustainability through higher customer satisfaction, while economic and environmental benefits depend on fixed costs and electricity emissions, respectively.
Cargo bikes	Load-dependent travel time is often simplified, but it strongly affects feasibility. Ignoring load and street gradient can increase costs or create infeasible solutions under time windows (Rave & Fontaine, 2025). Environmental benefits are automatic in home delivery as cargo bikes reduce van activity (Ozyavas et al., 2025).
Sustainability trade-off	ECN: ↑↓ ENV: ↑ SOC: ↓ Cargo bikes improve environmental and social outcomes in dense areas, but economic performance may decline when payload limits, labor time, and coordination costs are high.
Drones	Time and cost benefits are not universal. Drones can improve delivery speed, but their performance depends on coordination, drone capacity, battery endurance, fixed costs, and budget constraints (Chen et al., 2024a; Kitjacharoenchai et al., 2020). Environmental assessment remains limited. Drones are often described as battery-powered and less polluting, but several models mainly optimize time, cost, or latency rather than explicitly assessing emissions (Bruni et al., 2023; Kitjacharoenchai et al., 2020). Application feasibility remains limited. Drones are mainly discussed for rapid, urgent, remote, or small-parcel deliveries, but their use is constrained by payload capacity, flight range, battery endurance, and launch or recharge support (Bruni et al., 2023; Chen et al., 2024a; Kitjacharoenchai et al., 2020). Energy feasibility is often simplified. Many drone models use flight-time or distance limits, while load-dependent nonlinear energy use and uncertain flight times can affect feasibility and customer waiting time (Bruni et al., 2023).
Sustainability trade-off	ECN: ↑↓ ENV: ↑↓ SOC: ↑ Drones improve social sustainability by increasing delivery speed, but economic and environmental aspects may decline due to payload, energy efficiency, and fixed costs.
EVs	Full substitution remains limited. EVs can reduce emissions, but limited range, lower load capacity, high acquisition cost, and charging infrastructure constraints make all-EV deployment difficult (Juvvala & Sarmah, 2025; Wang et al., 2024). Charging strategy remains unresolved. En-route charging stations can extend EV operating range compared with depot-only charging, but they involve location, capacity, budget, and waiting-time constraints. Battery swapping can reduce service delays, but it requires additional infrastructure investment (Leng et al., 2023; Mirzavand Boroujeni et al., 2025; Yang et al., 2022). Energy and emission estimates are sensitive. Constant energy rates can misestimate EV performance, while payload, travel distance, travel time, ambient temperature, battery capacity, charging time, and dynamic–static emissions can change cost, feasibility, and environmental outcomes (Leng et al., 2023; Wang et al., 2024). Benefits depend on supporting nodes. EV performance improves when combined with PLs, mobile lockers, UCCs, waterways, or combined recharging and freight-replenishment facilities, but these integrations are still fragmented across separate models (Korkmaz et al., 2025; Mirzavand Boroujeni et al., 2025).
Sustainability trade-off	ECN: ↑↓ ENV: ↑ SOC: ↓ EVs improve environmental sustainability, but economic feasibility and social service reliability may decline under cost, range, charging, and payload constraints.
PLs	Environmental benefits are conditional. PLs can reduce delivery distance and failed home deliveries, but their environmental benefit depends on customer travel mode and locker accessibility. Cost-efficient locations may not create the lowest emission solution (Ozyavas et al., 2025). Cost–accessibility trade-off remains critical. Fewer large PLs may reduce operating costs, while many smaller PLs improve customer accessibility but can increase long-term costs (Ozyavas et al., 2025). Capacity and demand uncertainty are underdeveloped. Fixed PLs can face capacity shortages during peak demand, while mobile PLs improve flexibility but add relocation, replenishment, and routing decisions (Korkmaz et al., 2025). Integration with other delivery modes is not fully resolved. PLs are modeled with EVs, cargo bikes, mobile lockers, occasional drivers, or transfer-point functions, but these combinations are often studied separately (Enthoven et al., 2020; Mirzavand Boroujeni et al., 2025; Yu et al., 2021).
Sustainability trade-off	ECN: ↑↓ ENV: ↑↓ SOC: ↑ PLs improve social sustainability by lowering failed deliveries, but economic and environmental sustainability may decline due to opening costs and customer travel mode.

* ECN = Economic; ENV = Environmental; SOC = Social.

connectivity.

Generation 4: Network-wide node connectivity will be introduced. All nodes are linked through shared information systems, enabling coordinated planning, real-time updates, and collaborative LM-LRP designs. This provides the groundwork for reducing overall costs and increasing profits. Building on this interconnected structure, the final generation advances toward full automation and seamless collaboration across the network.

Generation 5: A fully automated and collaborative node ecosystem will be implemented. Parcel transfers, sorting, and last-mile delivery interactions will become automated across all nodes and modes. Continuous information flows on parcel status, delivery windows, and handling requirements will be available to all stakeholders, enabling a mature, synchronized logistics network.

6.2.2. *Logistics networks*

Generation 1: In the first generation of logistics networks, deliveries became more organized as carriers increasingly operated under time-window constraints. This period also saw the introduction of loading-zone booking systems that allowed carriers to reserve curbside spaces in advance. These systems reduced inefficiencies associated with searching for parking and queuing at busy locations. As these coordination mechanisms improved operational reliability, they also created conditions for resource sharing across carriers.

Generation 2: In the second generation, shared logistics became feasible through two reinforcing developments: the sharp rise in parcel volumes driven by e-commerce and the growing maturity of booking systems that improved the management of nodes and access points.

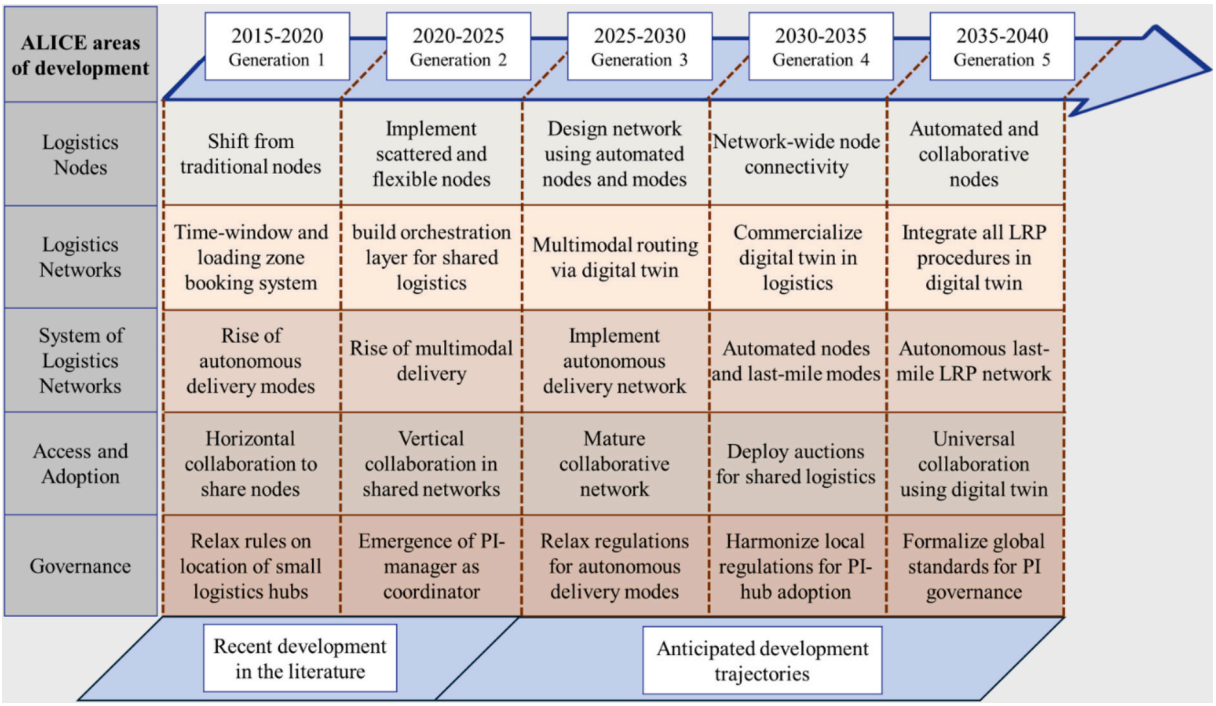


Fig. 8. Integrated framework for the strategic development of the LM-LRP.

These conditions made shared access to nodes particularly advantageous for smaller carriers that are highly cost-sensitive. However, increasing system complexity and service expectations require more flexible and integrated delivery configurations.

Generation 3: Although multimodal delivery has been explored in various studies, the third generation is expected to deliver a fully established multimodal LM-LRP. The emphasis shifts to combining automated modes, such as ADRs and drones, with existing modes, including vans, trucks, cargo bikes, and potentially public transport. Advances in battery and charging technologies, especially fast charging, support the operational integration of these modes. Because this multimodal LM-LRP introduces new interactions between modes and nodes, continuous asset and performance monitoring becomes essential.

Generation 4: In the fourth generation, vertical and horizontal collaboration will be expanded. A live representation of the LM-LRP, including inventory levels, real-time node capacities, the current status and location of delivery modes, and operational parameters such as battery state and maintenance requirements, must be captured. This progressive change necessitates a shift from booking systems to the digital twin.

Generation 5: By the fifth generation, all last-mile processes and stakeholders will be fully integrated, creating a transparent and equitable business ecosystem. At this stage, the digital twin reaches full maturity, enabling stakeholders to compare services, costs, and operational performance in real time. Equipped with advanced analytics and dynamic visualizations, the digital twin becomes the central orchestrator of the entire last-mile delivery network.

6.2.3. *System of logistics networks*

Generation 1: Although autonomous vehicles have been researched for decades, fully self-driving systems have not yet been commercialized, largely due to unresolved safety concerns. In the first generation, last-mile research therefore focused on smaller autonomous modes, particularly ADRs and drones. These modes were evaluated for their feasibility in replacing human couriers in the last leg and, in some cases, substituting for vans operating between intermediate and inner-echelon nodes. As these initial applications demonstrate feasibility at smaller scales, the next generation expands toward integrated delivery systems.

Generation 2: In the second generation, autonomous modes began to be combined with conventional vehicles, marking the rise of multimodal autonomous delivery. Studies explored a range of pairings, including truck–drone, van–drone, van–ADR, and even public transport–ADR combinations. These hybrid systems demonstrated the operational potential of coordinated human–autonomous delivery networks. However, scaling these hybrid systems requires more structured and dedicated autonomous network configurations.

Generation 3: The third generation is characterized by the implementation of an autonomous delivery network. ADRs and drones are expected to be widely deployed for deliveries to PLs, which are the only nodes capable of enabling automated parcel handover to customers without human involvement. While such applications already exist in countries such as China and Japan, their current scale is limited, and their long-term economic feasibility remains uncertain. Wider adoption and improved infrastructure can enhance efficiency and service reach, which are addressed in the next generation.

Generation 4: In the fourth generation, both nodes and last-mile modes are anticipated to become substantially automated. Parcel sorting, storage, and consolidation within nodes will require minimal human involvement, while ADRs and drones will dominate last-mile operations. Self-driving vans and trucks will continue to face regulatory constraints, restricting their role primarily to controlled environments. This progression sets the stage for the next generation, where automation is further expanded.

Generation 5: The fifth generation envisions a fully autonomous LM-LRP network. Advances in safety, precision, and public acceptance will enable automated vans and trucks to operate in urban areas and handle flows between outer, intermediate, and inner echelons. Transfers between all nodes and modes, not just PLs, will become automated, resulting in a seamless, end-to-end autonomous logistics network.

6.2.4. *Access and adoption*

Generation 1: Horizontal collaboration began to emerge, but it was limited to node sharing, where carriers rented space in shared hubs to reduce opening and operating costs. Over time, horizontal collaboration

expanded to the mode level, allowing carriers to use the same vehicle for deliveries, which reduced empty trips and lowered operational costs. This development highlights the need to extend horizontal collaboration toward coordination across different tiers.

Generation 2: In the second generation, vertical collaboration started to form as stakeholders at different tiers began coordinating their activities. Early examples included retailers sharing demand forecasts and inventory information with carriers to improve the alignment of replenishment flows and last-mile deliveries. This information exchange reduced stockouts, improved routing reliability, and helped minimize holding and transportation costs. As vertical collaboration developed further, operational coordination increased, such as synchronizing warehouse picking schedules with carrier dispatch times or allowing carriers to use retailer facilities as temporary cross-docking points. These developments point to the need for more integrated collaboration structures to support scaling.

Generation 3: The third generation represents the emergence of a mature collaborative network in which horizontal and vertical collaborations operate simultaneously. In such networks, carriers may even share customers to collectively meet the full demand of the service area at lower cost. While several studies have evaluated the benefits of fully collaborative systems, their real-world implementation requires an additional coordinating actor capable of managing relationships and ensuring fairness across stakeholders. This requirement for fairness is addressed in Generation 4 through auction-based allocation mechanisms.

Generation 4: In the fourth generation, open and shared networks create the foundation for fair and transparent pricing of LM-LRP procedures. Auction-based systems become necessary to allocate node capacity, vehicle space, or delivery tasks based on market-driven prices. Such bidding systems broaden participation by giving all companies the opportunity to contribute to last-mile fulfillment under equal competitive conditions. This creates the foundation for extending collaboration beyond local network boundaries.

Generation 5: In the fifth generation, collaboration and auction mechanisms extend beyond individual cities and become universally connected. A digital twin platform functions as an integrator across regions and stakeholder groups, enabling shared visibility, dynamic pricing, and real-time coordination. Although this research focuses on the last mile, universal collaboration through the digital twin connects global logistics with city logistics and reduces unnecessary back-and-forth movements. Goods can therefore be directed precisely to the locations where they are needed, improving efficiency, particularly in dense urban areas.

6.2.5. Governance

Generation 1: In several urban areas, the regulations related to establishing small transportation nodes are relaxed, allowing them to be located in residential land-use areas. This provides the opportunity to have micro-hubs and satellites in close proximity to customers. However, more frequent movement of transportation modes can cause noise or traffic that is not favorable for residents. These emerging externalities highlight the need for coordinated governance mechanisms.

Generation 2: The issues discussed in the first generation can worsen, especially considering the increase in collaborative distribution networks. To deal with this issue, a stakeholder that can coordinate shared assets is needed. The PI-manager, who oversees all nodes and modes that are shared and used by multiple stakeholders, is a promising solution. As coordination becomes more structured, regulatory challenges related to automation remain a key barrier.

Generation 3: While the adoption of automated nodes, such as PLs, is promoted by governments and policymakers, the opposite applies to automated modes. This is despite the fact that autonomous trucks have operated for nearly two decades with obstacle detection, high-resolution mapping, and autonomous navigation capabilities (Rees, 2012; TradeEarthmovers, 2018). This issue, rooted in conservative decision-making and institutional hesitation rather than technological immaturity, impedes wider automation in upstream layers. However, it is expected that, with advancements in technology, the acceptance of autonomous modes will increase, and they will dominate last-mile delivery. Building on this technological acceptance, Generation 4 focuses on stabilizing shared networks and hubs.

Generation 4: As shared networks mature, PI-hubs will dominate in the LM-LRP. Creating PI roadmaps and establishing PI working groups are among the activities supporting this area of development. These initiatives create the foundation for a fully integrated logistics system.

Generation 5: The ultimate development of the PI will be an integrated PI network in which all nodes and modes are shared and interconnected. This collaborative system enhances the efficiency and productivity of all assets, as they are used by multiple stakeholders.

7. Conclusion

In this study, the LM-LRP is systematically reviewed to address the lack of an integrated framework that facilitates the implementation of emerging technologies in the last mile. Emerging initiatives are classified as transportation nodes and modes and examined in terms of their network location, Tables 2 and 3, prevalence in the reviewed literature, Figs. 4 and 5, and applications across different network structures, Section 4. Fig. 6 illustrates these initiatives through a physical twin and digital twin structure and shows the flow of information between them. Mapping all identified components within the physical twin enables a comprehensive understanding of the interconnections among nodes, modes, and echelons, supporting their deployment within the network according to their functional roles. In contrast, the digital twin provides a unified environment that mirrors and synchronizes physical operations and facilitates the collaborative use of open and shared nodes and modes. Table 2 synthesizes key nodes, which are systematically reviewed and compared based on modeling implementation requirements, the most suitable echelons for establishment, and their advantages and disadvantages. A similar analysis is conducted for transportation modes, which are compared in terms of emissions, costs, load capacity, speed, and echelon suitability, as summarized in Table 3.

As the structure of the LM-LRP plays a critical role in the efficiency of the distribution network, an in-depth assessment of LM-LRP structures is provided in Section 4. Previous state-of-the-art studies are categorized into 1E-, 2E-, multi-, and adaptive-echelon LRP, and each study is extensively analyzed based on its objective function, employed technology, modeling application, solution methodology, and model functionalities. After that, in Section 5, the advantages of various node-mode combinations employed in the literature are examined, and an in-depth discussion of their main limitations and operational trade-offs is provided. This analysis is summarized in Table 8. Additionally, Fig. 7 visualizes how the identified initiatives support major LM-LRP objectives through operational enablers and performance targets.

The major research gaps in the LM-LRP literature are identified in Section 6. They include limited operational realism, fragmented technology integration, and insufficient governance and policy frameworks. The reviewed papers revealed that the impacts of emerging initiatives in the LM-LRP are not universally positive and depend on deployment context, infrastructure readiness, modeling assumptions, and operational constraints, as presented in Table 9. The table further shows that future LM-LRP research needs to move beyond isolated cost or emission objectives and adopt more integrated assessments of economic, environmental, and social performance. Addressing these gaps offers opportunities for impactful modeling and empirical research. Moreover, an actionable framework is introduced in Fig. 8 to identify opportunities for practical modeling and empirical research to address the identified gaps. This figure highlights both existing developments and the critical future actions required to advance toward a more integrated LM-LRP framework.

Among the identified gaps, improving the representation of real-

world operations should receive the highest priority. Many LM-LRP models still assume deterministic demand and travel times, which limits their practical relevance. Future studies should therefore develop stochastic and multi-period models and validate them using operational data and real-world case studies. Another priority is the integrated modeling of nodes and modes to identify the benefits of their combined application. Finally, research on shared networks should further examine data sharing, stakeholder incentives, benefit allocation, safety requirements, and the role of PI-managers.

This review has some limitations. From a methodological perspective, despite the rigor of the PRISMA 2020-based methodology, the exclusion of conference proceedings and non-English publications may introduce selection bias, and some relevant studies may be missing due to publication bias. Moreover, the Google Scholar search was restricted to title-only queries due to platform limitations, which may have reduced the identification of relevant studies. From a validation perspective, while the proposed forward-looking framework is grounded in the reviewed literature, it has not been validated through empirical case studies or expert consultation, which may limit its practical applicability.

CRediT authorship contribution statement

Masoud Kahalimoghadam: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Russell Thompson:** Supervision, Methodology, Funding acquisition, Conceptualization. **Michael Kirley:** Writing – review & editing, Visualization, Validation, Supervision, Funding acquisition, Conceptualization. **Lele Zhang:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A:. Solution Methodology

The evolution of LM-LRP methods has tracked rising complexity and practical orientation, since early work centered on exact MILP formulations solved by CPLEX or Gurobi that remain indispensable for proving optimality on small or structured instances and for setting benchmarks, as shown in single-echelon and early two-echelon studies and strengthened by compact reformulations and cutting planes (Ahmadi et al., 2015; Escobar-Vargas & Crainic, 2024; Gianessi et al., 2016; Singh et al., 2018), although exponential growth in variables and constraints limits purely exact models in realistic urban settings. Consequently the field shifted toward heuristics and metaheuristics, with ALNS becoming the main workhorse alongside tabu search, VNS, genetic algorithms, and constructive heuristics paired with Clarke–Wright savings, which exploit problem structure through destroy–and–repair operators, synchronization-aware neighborhoods, and hybrid fleet handling to scale to large and multi-period instances (Koç, 2016; Schiffer & Walther, 2018; Schneider & Löffler, 2017). Other metaheuristic approaches include tabu search, VNS, and genetic algorithms (Anderluh et al., 2021; Breunig et al., 2016; Nguyen et al., 2012; Simoni et al., 2018; Zhao et al., 2018). A subsequent wave embraced matheuristics and hybrids that couple exact modules with metaheuristic search, including column generation with the elementary shortest path problem with resource constraints as a subproblem, Lagrangian relaxation with ADMM, branch-and-check with tailored cuts, and set-partitioning linked to VRP formulations (Bruni et al., 2023; Ghiani et al., 2025; Liu et al., 2023; Yang et al., 2022). Exact subproblem solvers secure feasibility and strong bounds, while metaheuristics explore large neighborhoods efficiently.

Method selection also follows network coupling, because in 1E-LRPs compact MILPs still serve as references for small or structured instances while scalability is typically achieved through ALNS and related metaheuristics or through matheuristics that embed column generation or Lagrangian relaxation with ADMM, whereas in two- and multi-echelon settings the added cross-level flow conservation, transfer capacities, and synchronization constraints favor structured time–space formulations with dynamic discretization for exact methods and, in practice, echelon-wise decomposition, synchronization-aware neighborhoods, iterative coordination, and continuous approximation for first-echelon districting with hybrid embedding of second-echelon routing, particularly when satellites and PLs are mobile or temporary and feasibility hinges on time–space alignment (Korkmaz et al., 2025; Tian & Hu, 2023).

More recently, learning-based and hyper-heuristic approaches have introduced self-adaptation in operator choice and search control, for example a Q-learning hyper-heuristic within simulated annealing to manage heterogeneous EV fleets under temperature-dependent energy use (Leng et al., 2023) and adaptive ALNS for van–robot pickup-and-delivery with synchronization between modes (Yu et al., 2022). In parallel, simulation and approximation have been integrated to increase realism and scalability under uncertainty, with continuous approximation supporting urban districting and transfer design (Arevalo-Ascanio et al., 2024; Janjevic et al., 2019). Furthermore, probabilistic capacity models capture stochastic demand and travel times in last-mile systems, yet city-scale empirical validation remains limited to date (Ozyavas et al., 2025; Zhao et al., 2025).

A consistent methodological trajectory can be discerned. Exact approaches provide theoretical rigor and guarantee optimality but are confined to small or highly structured problem instances. Metaheuristics, by contrast, enable scalability to realistic problem sizes, albeit at the expense of optimality guarantees. Matheuristics offer an intermediate balance by leveraging decomposition techniques to integrate the strengths of exact optimization with the flexibility of heuristic search. More recently, learning-based approaches have emerged, showing promise for adaptive scalability; however, their validation remains limited. Future research directions include: (i) the development of scalable, uncertainty-aware algorithms that unify stochastic and robust optimization paradigms (Guo et al., 2024); (ii) the systematic integration of sustainability metrics into heuristic frameworks, rather than addressing them as secondary objectives (Du et al., 2023; Su et al., 2025); and (iii) the empirical validation of hybrid approaches using real-world operational data, thereby strengthening methodological credibility beyond computational experimentation (Rautela et al., 2022; Wang et al., 2023).

Exact methods

This subsection reviews the use of exact optimization methods in the LM-LRP, principally those implemented using commercial solvers such as IBM

ILOG CPLEX and Gurobi. It includes both standalone exact formulations and exact components embedded within hybrid solution frameworks, particularly where mixed-integer programming (MIP) or MILP models are solved exactly for small instances or specific subproblems. Table A1 summarizes studies that use these solvers as exact methods, and mapping solver use to network structure indicates that more than seventy percent of two-echelon LRP studies rely on commercial solvers, which are applied across networks with PLs, satellites, UCCs and micro-hubs, drones, autonomous robots, and cargo bikes, while a similar pattern appears for functional extensions, since models with synchronization, sustainability, pickup and delivery, stochastic elements, and dynamic hub location are also commonly handled by commercial solvers.

Table A1
Studies using commercial solvers within exact approaches.

Reference	Methodology	Node / Mode	Network	Function
Ahmadi et al. (2015)	Exact: two-stage stochastic MILP solved to optimality (GAMS / CPLEX).	NA	2E-LRP	Stochastic; Uncertainty
Singh et al. (2018)	Exact: Formulated as MILP, solved using IBM ILOG CPLEX 12.5.	NA	1E-LRP	NA
Gianessi et al. (2016)	Exact: MIP solved with the commercial solver CPLEX	NA	2E-LRP	NA
Aloui et al. (2021)	Hybrid – Exact MILP (CPLEX) for small instances; heuristics for larger.	NA	2E-LRP	Sustainability; Synchronization
Ouhader and El Kyal (2023)	Exact: MILP solved via commercial solver (B&B in MATLAB/CPLEX) + ε-constraint for Pareto sampling.	UCCs	2E-VRP	Sustainability; Synchronization
Yu et al. (2020)	Hybrid – Exact: MIP solved by CPLEX for small instances; heuristics for larger.	Satellites	2E-LRP	NA
Zhao et al. (2018)	Hybrid – Exact MIP (Gurobi) for small instances; heuristics for larger.	NA	2E-LRP	NA
Chen et al. (2024a)	Exact: MILP formulation solved by Gurobi (branch-and-cut) for small instances; heuristics for large.	Drones/Hybrid modes	2E-LRP	NA
Rautela et al. (2022)	Exact: MILP combined with continuum approximation; solved by Gurobi.	NA	2E-LRP	Pickup and delivery
Amini and Haughton (2023)	Exact: scenario-based MILP (explicit scenarios for customer presence).	NA	2E-LRP	Stochastic
Guo et al. (2024)	Exact, as a hybrid solution, solved by CPLEX 12.8	Micro-hubs	2E-LRP	Dynamic hub location
Kitjacharoenchai et al. (2020)	Hybrid: MIP (GAMS/CPLEX) for small instances + LNS/metaheuristic for larger instances.	Drones/Hybrid modes	2E-LRP	Synchronization
Korkmaz et al. (2025)	Formulated as integer linear programming and solved with the commercial IBM OPL / CPLEX	PL/ EV/ drones	2E-LRP	Synchronization; Dynamic hub location
Enthoven et al. (2020)	CPLEX 12.8	PL; Cargo bike	2E-VRP	NA
Liu et al. (2022)	MILP formulation solved with Gurobi.	Micro-hubs; ADRs	2E-VRP	NA
Zhou et al. (2024)	CPLEX 12.9	Satellites	2E-VRP	NA
Janjevic et al. (2019)	Solved by a numerical solver (Gurobi 6.5.1).	NA	Multi-echelon LRP	NA

Within the exact-solution literature for the LM-LRP, branch-and-price appears more frequently than branch-and-check, since both are embedded in branch-and-bound yet tackle the linear-programming relaxation in different ways, with branch-and-price addressing formulations that contain an excessive number of variables by solving node relaxations through column generation in which a pricing subproblem introduces only variables with negative reduced cost and thereby strengthens bounds without enumerating the full variable set, whereas branch-and-check suits formulations with an intractable or implicitly defined constraint set by solving a compact relaxation and then subjecting incumbent solutions to a feasibility check that adds a violated lazy constraint whenever it is detected and continues the search. Table A2 synthesizes studies that employ branch-and-price or branch-and-check for the LM-LRP, and two-echelon networks predominate over one-echelon designs while node and mode selections remain diffuse with recurring use of EVs, drones, and satellites, and functional orientations are broadly balanced across pickup and delivery, dynamic hub location, stochastic modeling, and synchronization.

Table A2
Studies that apply branch-and-price or branch-and-check as exact solution methods.

Reference	Methodology	Node / Mode	Network	Function
Liu et al. (2023)	Branch-and-price / Column-generation	Mobile PL	1E-LRP	Stochastic
Ozyavas et al. (2025)	Branch-and-price / Column-generation	Cargo bikes; PL	2E-LRP	Pickup and delivery; dynamic hub location; Stochastic
Tian and Hu (2023)	Branch-and-price / Column-generation	Satellites/ Drones	2E-LRP	dynamic hub location; Synchronization
Wang et al. (2022)	Branch-and-price with column generation	EVs	2E-LRP	Pickup and delivery
Bruni et al. (2023)	Branch-and-Check with custom no-good cuts	Drones/ Hybrid modes	1E-LRP	Stochastic; Uncertainty
Pourmohammad-Zia and van Koningsveld (2024)	Branch-and-price (Dantzig–Wolfe decomposition) for first-echelon	Satellite	2E-LRP	Synchronization

Another exact identified method is dynamic discretization discovery applied by Escobar-Vargas and Crainic (2024) to solve the two-echelon multi-attribute LRP with synchronization. The problem is formulated as a mixed-integer program and addressed through an iterative refinement of a reduced time–space network until optimality is achieved, thus avoiding the combinatorial explosion of a fully discretized time–space model.

Heuristics, metaheuristics, and matheuristics

Adaptive large neighborhood search (ALNS) is a destroy-and-repair metaheuristic widely applied in transportation planning and the LM-LRP because it scales efficiently to large instances and accommodates rich, real-world constraints. By adaptively selecting among multiple removal and insertion operators based on recent performance, ALNS balances diversification and intensification and performs effectively in multi-echelon settings, synchronization, time-window management, pickup-and-delivery operations, and electrified fleets. Within the LM-LRP literature, it appears both as a standalone solver and embedded in hybrid metaheuristics, for example in combination with MILP, dynamic programming, or column generation, and is frequently tailored to second-echelon route improvement in two-echelon designs. Recent applications further integrate drones, PLs, cargo bikes, and micro-hubs, increasingly addressing joint objectives such as cost efficiency and sustainability.

As shown in Table A3, ALNS is used predominantly in two-echelon contexts, typically framed as 2E-VRP or 2E-LRP, with comparatively fewer single-echelon applications. Node and mode choices are diverse, spanning electric vehicles, drones, PLs, cargo bikes, micro-hubs, and occasional satellites, while some studies leave the mode unspecified. Functional emphases concentrate on synchronization, with sustainability and pickup-and-delivery also represented, and a smaller set of studies reporting no explicit functional extension. Methodologically, ALNS is employed both independently and in hybrid designs, for example in conjunction with MILP, dynamic programming, ant colony optimization, or simulated annealing acceptance, and is in some cases restricted to second-echelon route improvement.

Table A3
Articles using LNS and ALNS (solely or as part of a hybrid approach) to solve the LM-LRP.

Reference	Methodology	Node / Mode	Network	Function
Hemmelmayr et al. (2012)	ALNS	NA	2E-VRP	NA
Breunig et al. (2016)	LNS	Satellite	2E-VRP	Synchronization
Koç (2016)	ALNS; MILP formulation	NA	1E-LRP	NA
Koç et al. (2016)	ALNS		1E-LRP	Sustainability
Schiffer & Walther (2018)	ALNS + dynamic programming	EVs	1E-LRP	Sustainability
Kitjacharoenchai et al. (2020)	LNS	Drones	2E-LRP	Synchronization
Enthoven et al. (2020)	ALNS	PLs; Cargo bike	2E-VRP	NA
Anderluh et al. (2021)	ALNS	EVs	2E-VRP	Synchronization; Sustainability
Yu et al. (2021)	ALNS	PL	2E-VRP	Synchronization
Yu et al. (2022)	ALNS	ADRs	2E-VRP	Synchronization
Zhou et al. (2024)	ALNS	NA	2E-VRP	Synchronization
Guo et al. (2024)	ALNS	Micro-hubs	ALNS for route improvement	NA
Chen et al. (2024b)	ALNS		2E-LRP	Pickup and delivery
Chen et al. (2024a)	ALNS combined with ant colony	Drones	2E-LRP	
Pourmohammad-Zia and van Koningsveld (2024)	ALNS (destroy-and-repair operators, local search, k-means clustering) for second-echelon	EVs	2E-LRP	Synchronization
Rave & Fontaine (2025)	ALNS with tactical allocation split	Cargo bikes; micro-hubs	1E-LRP	
Mirzavand Boroujeni et al. (2025)	LNS with simulated-annealing acceptance	EVs; PLs; Hybrid modes	1E-LRP	

The other heuristic solution is Clarke and Wright savings algorithm which has been widely adapted for the LM-LRP, where routes are constructed once depots, satellites, or hubs are fixed and constraints on capacity, time windows, and service are enforced. Joint location–routing variants incorporate opening, assignment, and handling costs so facilities and routes are optimized simultaneously, while in two-echelon systems the method is applied sequentially, first to connect depots and satellites and then to design routes within each satellite with load balancing across levels. Randomized and hybrid extensions diversify construction and refine solutions with local search or matheuristics, providing strong starting points for metaheuristics such as ALNS and simulated annealing. As shown in Table A4, Clarke and Wright is used mainly in 2E-LRP, with one multi-echelon case and no clear single-echelon examples. Node and mode choices include PLs, EVs, and satellites, with some studies not specifying a mode. Functional aims span pickup and delivery, synchronization, sustainability, and stochastic formulations, and one study combines Clarke and Wright with VNS in a hybrid design.

Table A4
Articles using Clarke & Wright Savings (solely or as part of a hybrid approach) to solve the LM-LRP.

Reference	Methodology	Node / Mode	Network	Function
Li et al. (2021a)	Clarke & Wright	PL	2E-LRP	Pickup and delivery; Synchronization; Sustainability
Zhao et al. (2018)	Clarke & Wright	NA	2E-LRP	NA
Nguyen et al. (2012)	Clarke & Wright	NA	2E-LRP	NA
Wang et al. (2023)	Clarke & Wright savings and VNS	EVs	2E-LRP	Pickup and delivery
Dai et al. (2019)	Clarke & Wright	Satellites	Multi-echelon LRP	Stochastic

Other heuristic algorithms are shown in Table A5. Most heuristic contributions address 2E-LRP, with fewer studies on 2E-VRP and a small set on multi echelon designs. Node and mode choices are varied, including EVs, drones or ADRs, PLs, satellites, micro-hubs, and some papers leave modes unspecified. Functionally, synchronization and dynamic hub location recur most often, with sustainability and stochastic elements also prominent. Methodologically, the set spans constructive and greedy and clustering based searches, VNS and iterative greedy, immune algorithms often paired with NSGA-II, particle swarm optimization with differential evolution, decomposition with iterative integer linear programming for scalability, and bespoke schemes such as sequential route extraction or slime mold inspired metaheuristics.

Metaheuristics offer general purpose frameworks when exact methods become impractical on large instances, since genetic algorithms, simulated annealing, tabu search, ant colony optimization, particle swarm optimization, and adaptive LNS explore the solution space through stochastic operators with diversification and intensification and iterative improvement, and are frequently hybridized with local search such as 2-opt, relocate, and swap to raise solution quality and accelerate convergence, whose flexibility and scalability enable treatment of multi-echelon networks, time windows, heterogeneous fleets, and environmental objectives.

Table A5
Other heuristic methods applied to solve the LM-LRP.

Reference	Methodology	Node / Mode	Network	Function
Korkmaz et al. (2025)	Two-stage decomposition and iterative integer linear programming (relax & re-solve) for larger cases	PLs; Drones	2E-LRP	Synchronization; Dynamic hub location
Sutrisno and Yang (2023)	Clustering-based simultaneous neighborhood search	Satellite	2E-LRP	Synchronization
Du et al. (2023)	Hybrid Immune Algorithm and improved NSGA-II	EVs	2E-LRP	Sustainability
Liu et al. (2021)	Artificial immune algorithm and NSGA-II	Drones/Robots; dynamic satellites/PLs	2E-LRP	Synchronization
Tian and Hu (2023)	Sequential route extraction, farthest-insertion	Satellites/Drones	2E-LRP	dynamic hub location; Synchronization
Ahmadi et al. (2015)	VNS	NA	2E-LRP	Stochastic; Uncertainty
Guo et al. (2024)	Modified sweep for initialization; iterative greedy	Micro-hubs	2E-LRP	Dynamic hub location
Juvvala and Sarmah (2025)	Particle Swarm Optimization with Differential Evolution	EVs	2E-LRP	Sustainability;Stochastic
Liu et al. (2022)	Clustering-based artificial immune algorithm	Micro-hubs; ADRs	2E-VRP	NA
Janjevic et al. (2019)	Greedy/constructive heuristics and reduced enumeration	NA	Multi-echelon LRP	NA
Wang et al. (2024)	Optics temporal-spatial discretization combined with multi-strategy slime mould algorithm	EVs	Multi-echelon LRP	Sustainability
Arevalo-Ascanio et al. (2024)	Continuous multi-scale approximation and districting	NA	Multi-echelon LRP	Stochastic; Uncertainty

Matheuristics combine this exploratory power with the rigor of mathematical programming, since LM-LRP implementations often embed MILP subproblems within a metaheuristic framework or use metaheuristics to produce high-quality starting solutions for exact solvers, with examples that include LNS supported by integer-programming repairs, decompositions where routing is handled heuristically while location is optimized via MILP, and branch-and-price integrated with metaheuristic components, which exploit problem structure while maintaining scalability and typically improve accuracy and robustness relative to pure metaheuristics, although without full optimality guarantees.

Table A6 synthesizes metaheuristic contributions that concentrate on 2E-LRP with some single-echelon cases, covering VNS with diverse shaking and local-search schemes, genetic and memetic algorithms, ALNS combined with ant colony optimization, and hybrid multiobjective particle swarm optimization often seeded by constructive routines such as Clarke–Wright savings or k-means clustering, with applications that span drones and other hybrid platforms as well as EVs, and with functional aims that commonly include sustainability, pickup and delivery, stochastic modeling, and synchronization.

Table A7 surveys matheuristic studies that couple mathematical programming structure with heuristic search to achieve scalability without sacrificing model fidelity. Examples include Lagrangian relaxation with ADMM and dynamic programming subproblems, two-step set-partitioning or location-based formulations followed by traveling salesman problem or VRP route construction, and the hybrid method proposed by Gianessi et al. (2016), which combines nearest-neighbor construction with 2-opt improvement. These works cover both single- and 2E-LRP, often in settings with EVs or alternative delivery resources, and they emphasize pickup and delivery and sustainability, with other features left unspecified in some cases.

Table A6
Studies that apply metaheuristic methods to solve the LM-LRP.

Reference	Methodology	Node / Mode	Network	Function
Nguyen et al. (2012)	Clarke and Wright	NA	2E-LRP	NA
Ahmadi et al. (2015)	VNS, including shaking (insertion, swap, Or-opt, inverse Or-opt, depot opener/closer) and local search (2-opt, 3-opt, add/drop operators)	NA	2E-LRP	Stochastic; Uncertainty
Simoni et al. (2018)	Hybrid GA/memetic algorithm	NA	2E-LRP	Sustainability
Chen et al. (2024a)	Multi-start improved ALNS combined with ant colony	Drones/Hybrid modes	2E-LRP	NA
Wang et al. (2023)	Three dimensional k-means clustering for initialization; hybrid multi-objective particle swarm optimization with Clarke and Wright savings and VNS	NA	2E-LRP	Pickup and delivery
Su et al. (2025)	NSGA-II (Pareto front)	NA	1E-LRP	Sustainability
Zhao et al. (2025)	Two-stage hybrid Simulated Annealing — Attention Model	PLs	2E-LRP	Stochastic

(*continued on next page*)

Table A6 (*continued*)

Reference	Methodology	Node / Mode	Network	Function
Pourmohammad-Zia and van Koningsveld (2024)	ALNS (destroy-and-repair operators, local search, k-means clustering) for second-echelon	EVs	2E-LRP	Synchronization
Zhou et al. (2017)	Genetic algorithm and simulated annealing for bilevel MILP	NA	2E-LRP	Stochastic
Aloui et al. (2021)	Genetic algorithm and modified k-means clustering	NA	2E-LRP	Sustainability; Synchronization
Kır and Comert (2024)	Ant colony optimization	NA	2E-VRP	Pickup and delivery
Yu et al. (2020)	Simulated annealing	Satellites	2E-LRP	NA
Leng et al. (2023)	Simulated annealing as a part of Q-learning hyper-heuristic	EV	1E-LRP	Sustainability and pickup-and-delivery

Table A7
Studies applying matheuristic methods to the LM-LRP.

Reference	Methodology	Node / Mode	Network	Function
Yang et al. (2022)	Matheuristic: Lagrangian Relaxation and ADMM with dynamic programming for subproblems.	EVs	1E-LRP	Pickup and delivery
Ghiani et al. (2025)	Two-step hybrid set-partitioning and location-based model with location based heuristic, followed by traveling salesman problem/VRP route construction	ADRs	1E-LRP	Sustainability
Gianessi et al. (2016)	Matheuristic: Combining a nearest neighbor with 2-opt heuristic	NA	2E-LRP	NA

Appendix B

The full list of journals that are grouped under "Others" in Fig. 3 is listed below.

a) Applied Soft Computing;
b) Case Studies on Transport Policy;
c) Environmental Science and Pollution Research;
d) International Journal of Disaster Risk Reduction;
e) International Journal of Industrial Engineering Computations;
f) International Journal of Systems Science: Operations & Logistics;
g) International Transactions in Operational Research;
h) Journal of Cleaner Production;
i) Journal of Industrial and Production Engineering;
j) Networks;
k) Omega;
l) Operations and Supply Chain Management: An International Journal;
m) Research in Transportation Business & Management;
n) Supply Chain Analytics;
o) Transportation Letters;
p) Transportation Research Part B: Methodological;
q) Transportation Research Record;
r) International Journal of Production Research.

Data availability

Data will be made available on request.

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